Contents

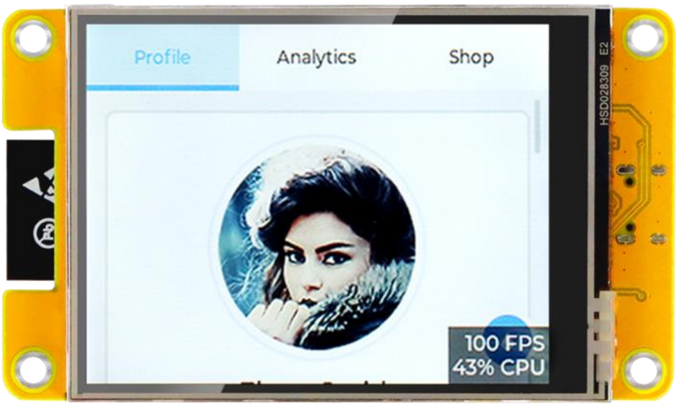
Preface

Important Notes

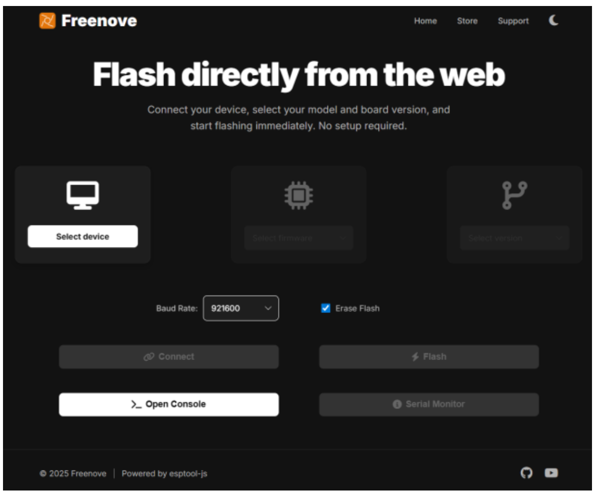
When you first power on the Freenove ESP32 Display, the screen will display a dynamic demo interface, featuring a modern user profile UI.

Please rest assured! This is not "malicious software" or a "used device."

This is a classic **benchmark/demo program** built on LVGL, a popular open-source embedded graphics library. It serves as an industry-standard tool to evaluate and demonstrate the development board's performance in rendering smooth graphics, handling animations, and supporting responsive touch interactions. You can easily overwrite it by **uploading a new sketch**. When you follow the tutorial to upload your first program, this demo will be automatically erased and replaced.



All models of the Freenove ESP32 Display are compatible with [NerdMiner] and [NMMiner]. Although the device does not come with the firmware pre-installed, there's no need to worry. We provide a convenient **web-based online flashing tool** and **a complete flashing tutorial**. There's no need to download any complex software. You can easily complete the flashing process in just a few minutes by selecting the firmware that matches your device model.



If you have any concerns, please feel free to contact us via support@freenove.com

Miner

This tutorial aims to introduce two excellent Bitcoin Mining projects. Freenove has adapted and validated these projects to ensure stable operation on our Freenove ESP32 Display, and have accordingly compiled detailed usage tutorials.

Disclaimer:
Freenove is not the original developer of these projects. All development, maintenance, feature implementation, and commercial licensing are the sole responsibility of their respective official developers.

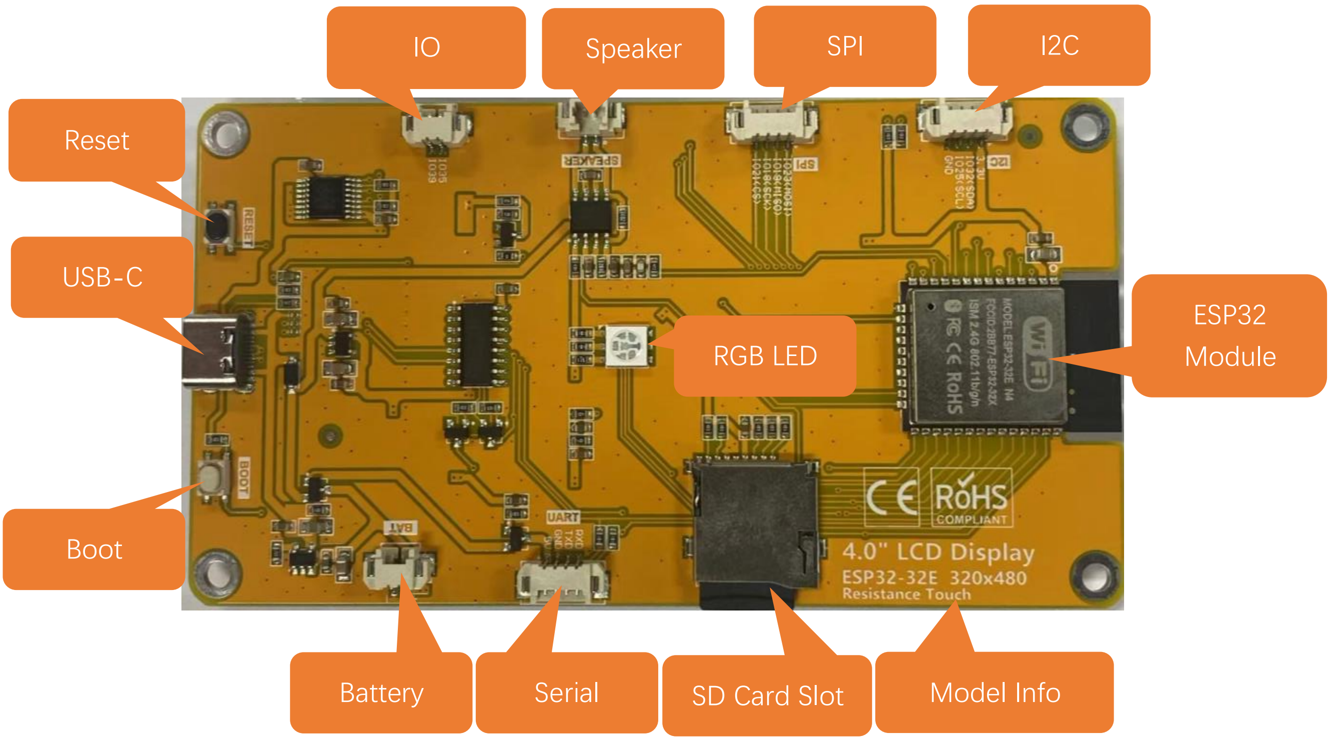
Miner Compatibility

Provide a tutorial to show how to install the following miner firmware.

Model	NMMiner (2.99 USD per device)	NerdMiner (free)
2.8 Inch ST7789 TN	 fully compatible	 fully compatible
2.8 Inch ILI9341 TN	 fully compatible	 fully compatible
3.2 Inch ST7789 IPS	 fully compatible, light theme	 fully compatible
3.5 Inch ST7796 TN	 fully compatible	 fully compatible
4.0 Inch ST7796 TN	 fully compatible	 fully compatible

Freenove ESP32 Display

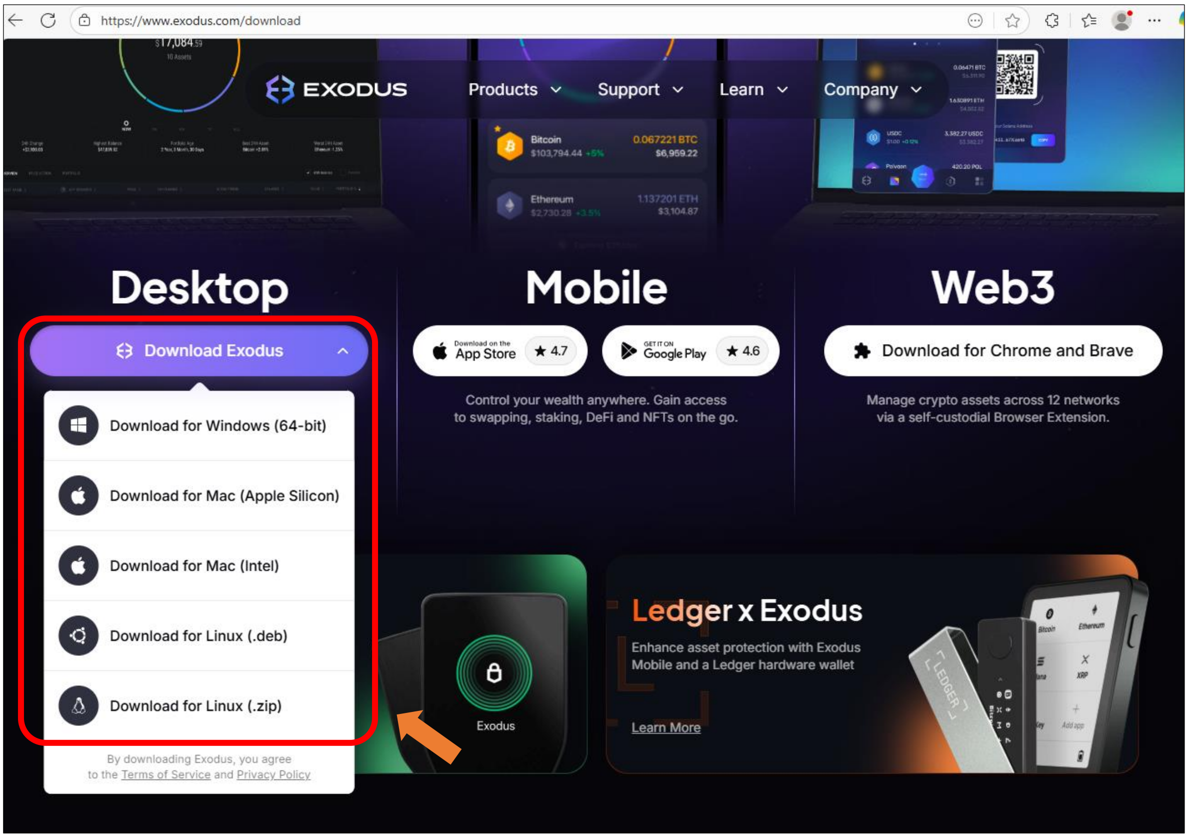
Hardware Interfaces



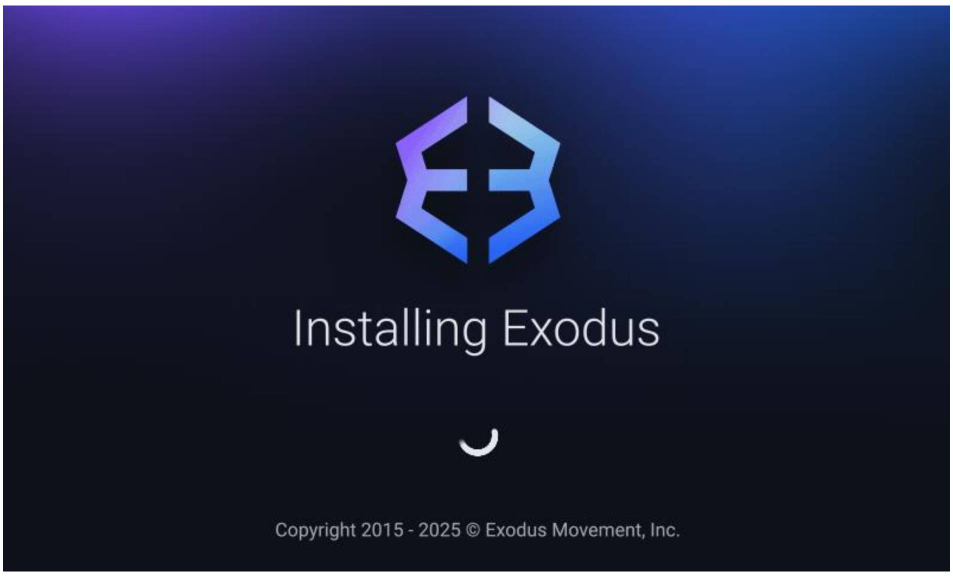
Obtaining the BTC Receiving Address

Note: If you already have a BTC receiving address, feel free to skip this step. Otherwise, the following section will guide you through obtaining a new one.

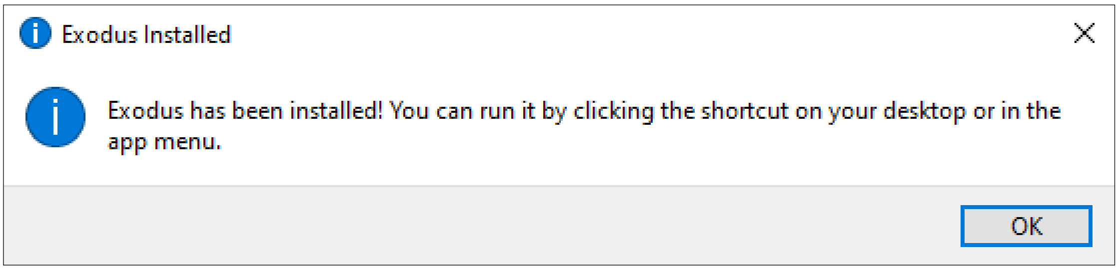
1. Click this link (https://www.exodus.com/download) to open the Exodus downloading interface. Choose the version consistent with your system.



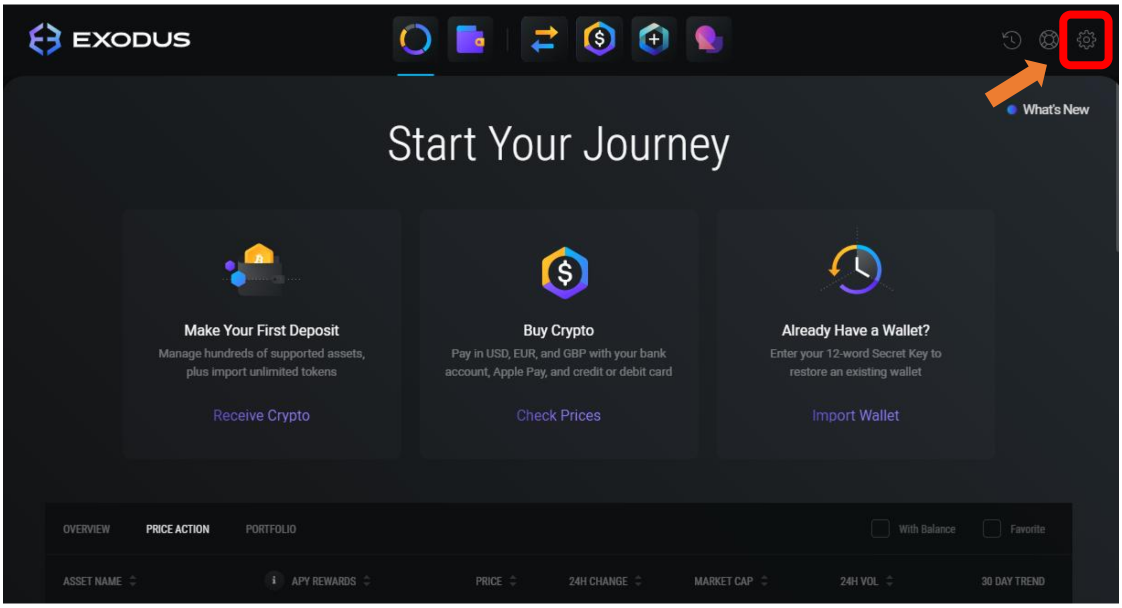
2. Open the downloaded installation package to install.



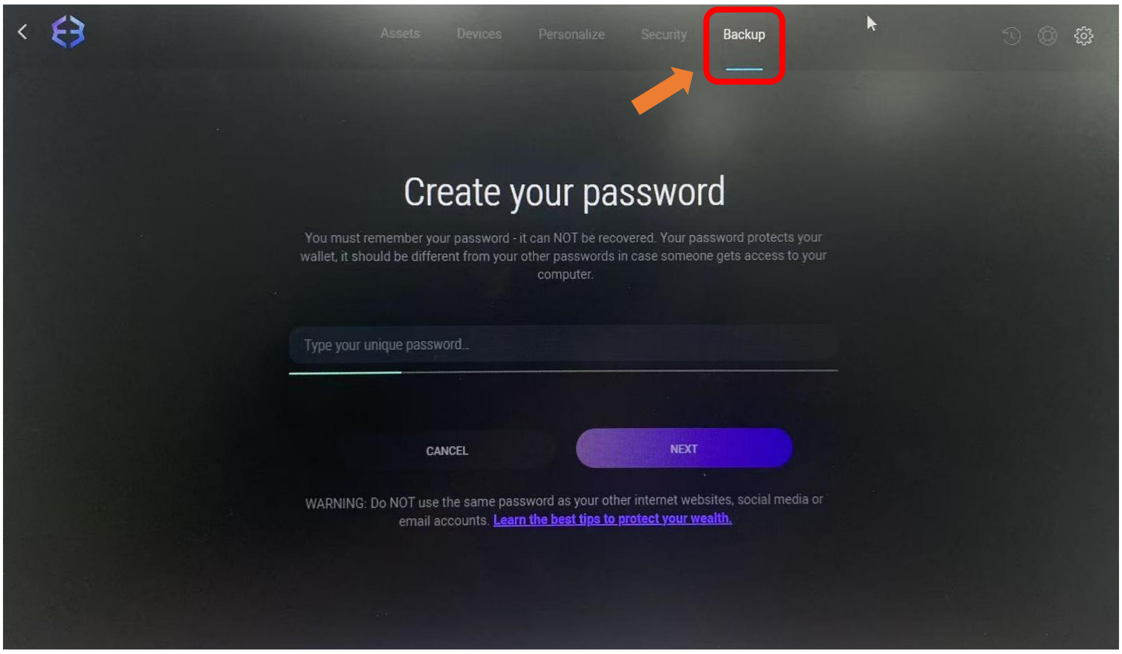
3. The following prompt indicates that the installation is successful.



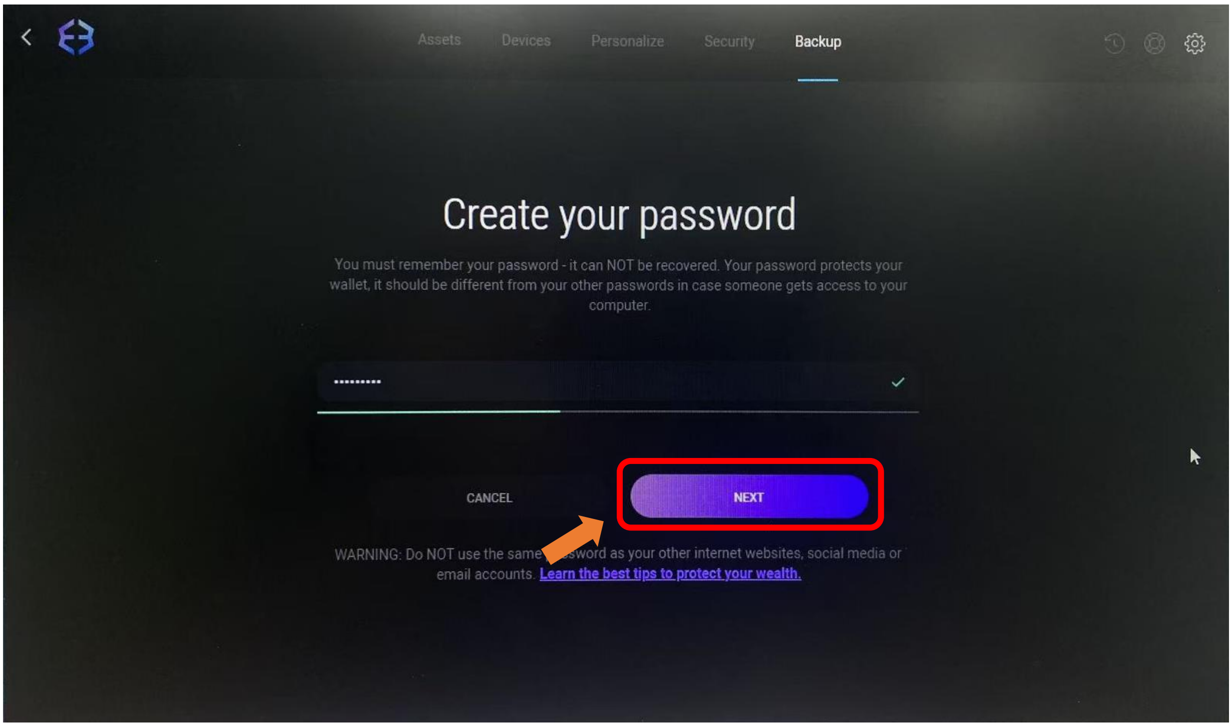
4. Open **Exodus**, click "**Setting**".



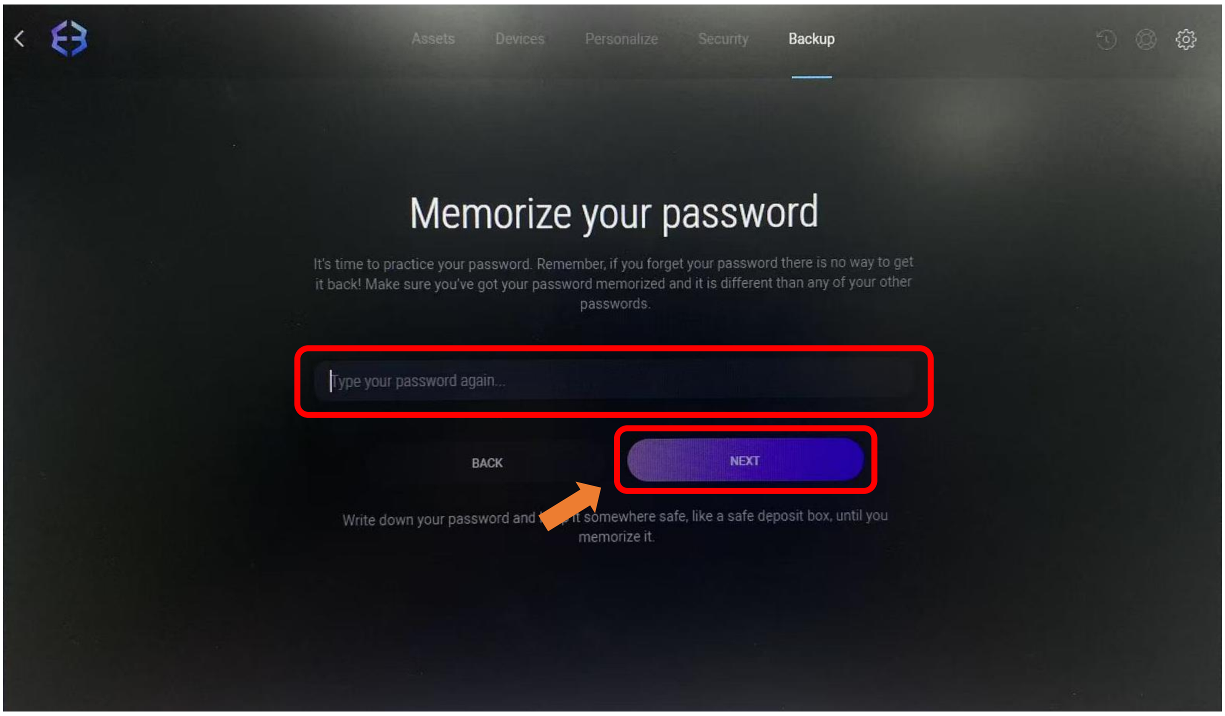
5. Click "**Backup**" to create your password for registering your wallet.
Important Note: Do not forget or share this password with anyone.



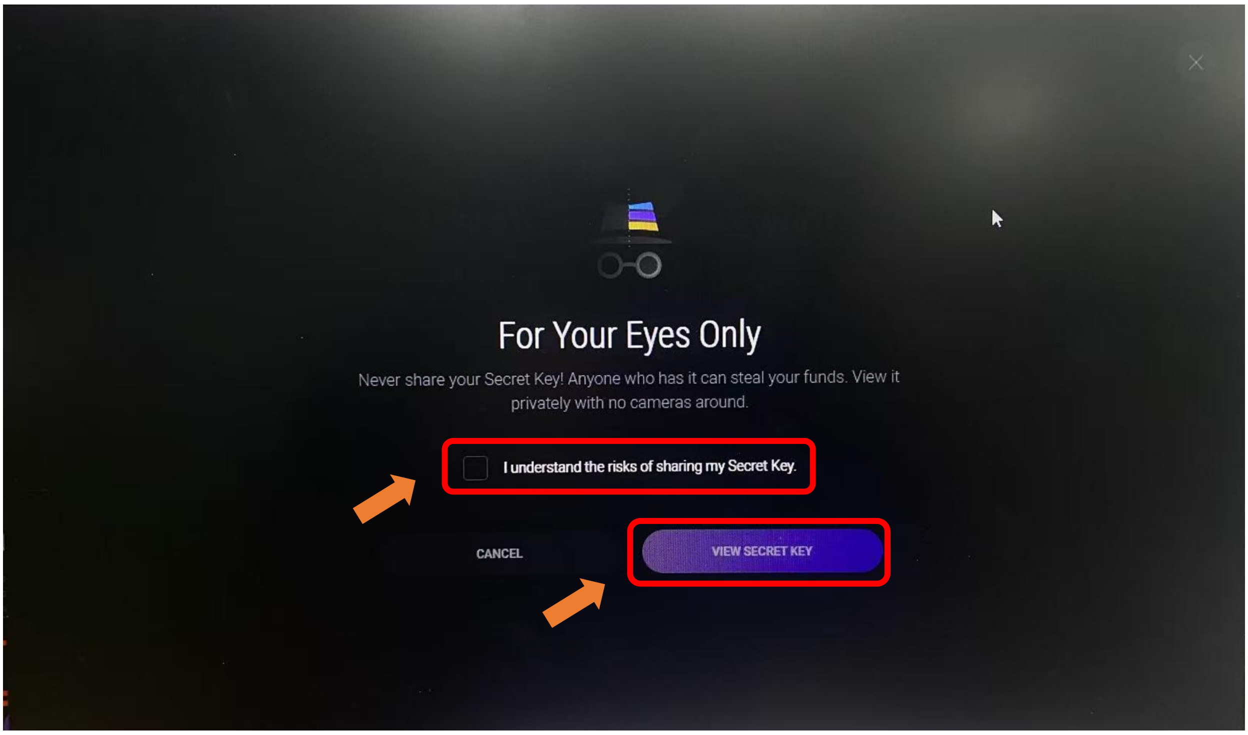
6. Click "**NEXT**".



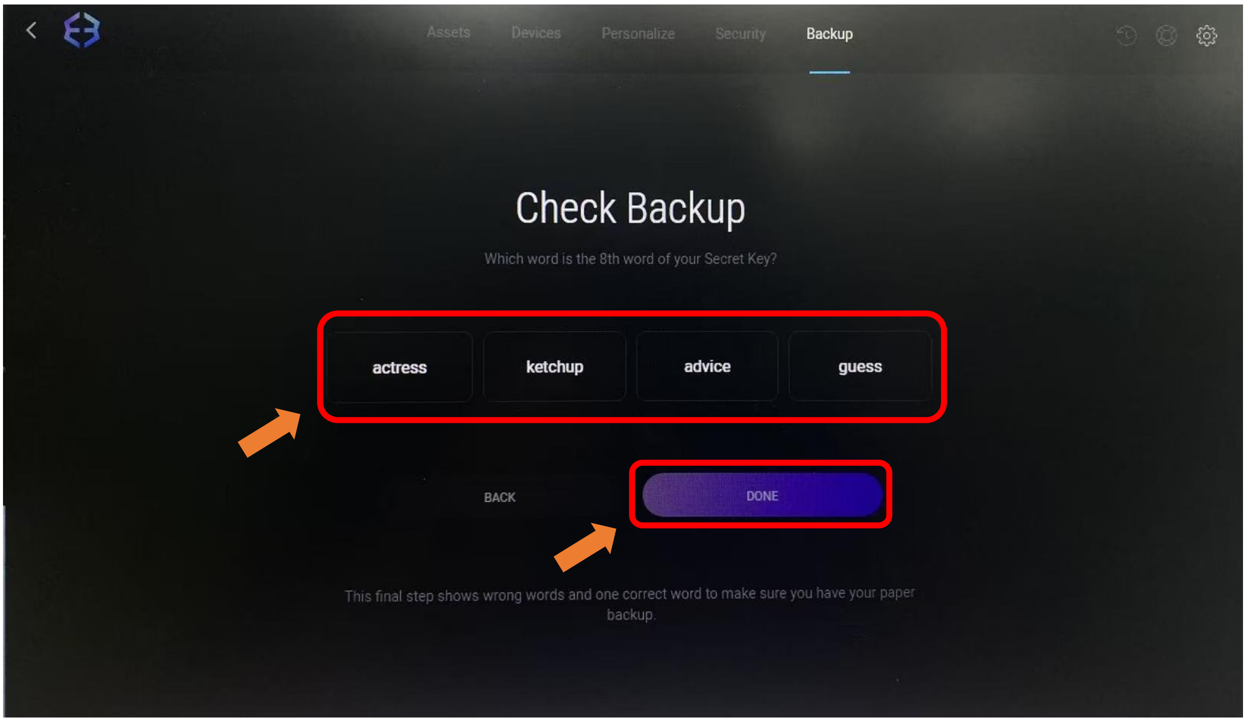
7. Enter the password again and click "**NEXT**".



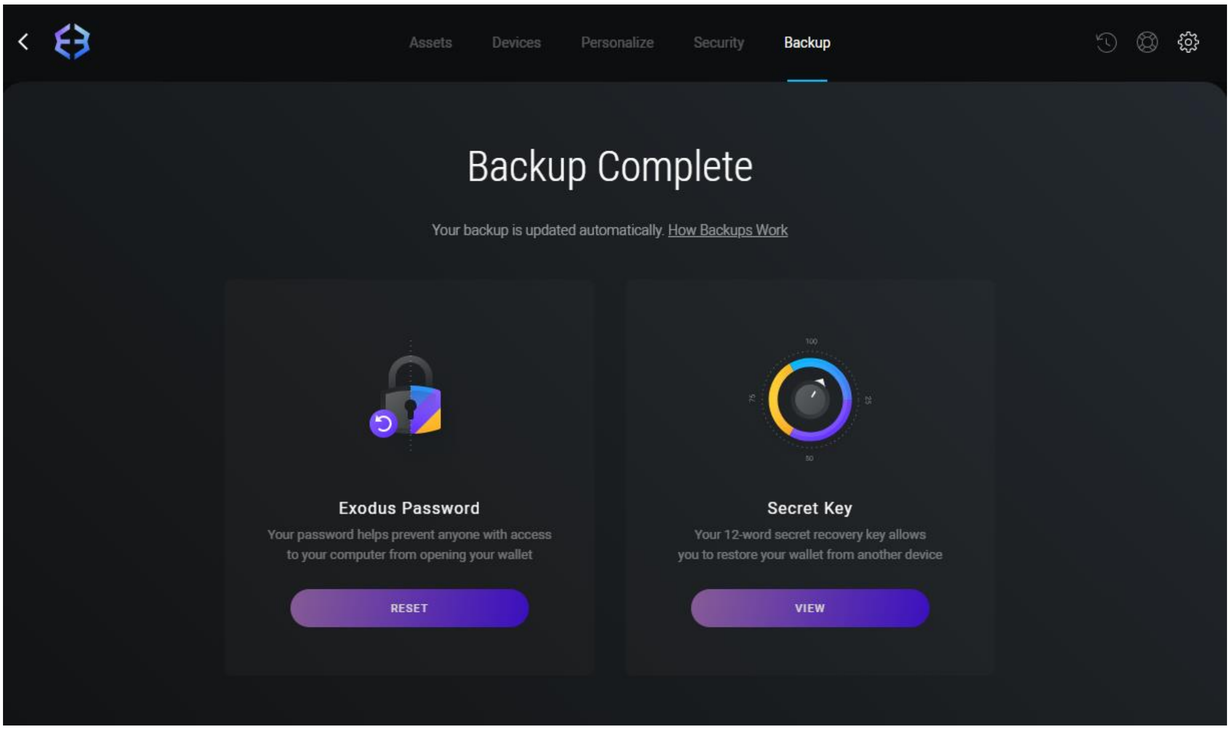
8. Check the box “**I understand the risks of sharing my Secret Key**”, and click “**VIEW SECRET KEY**”.



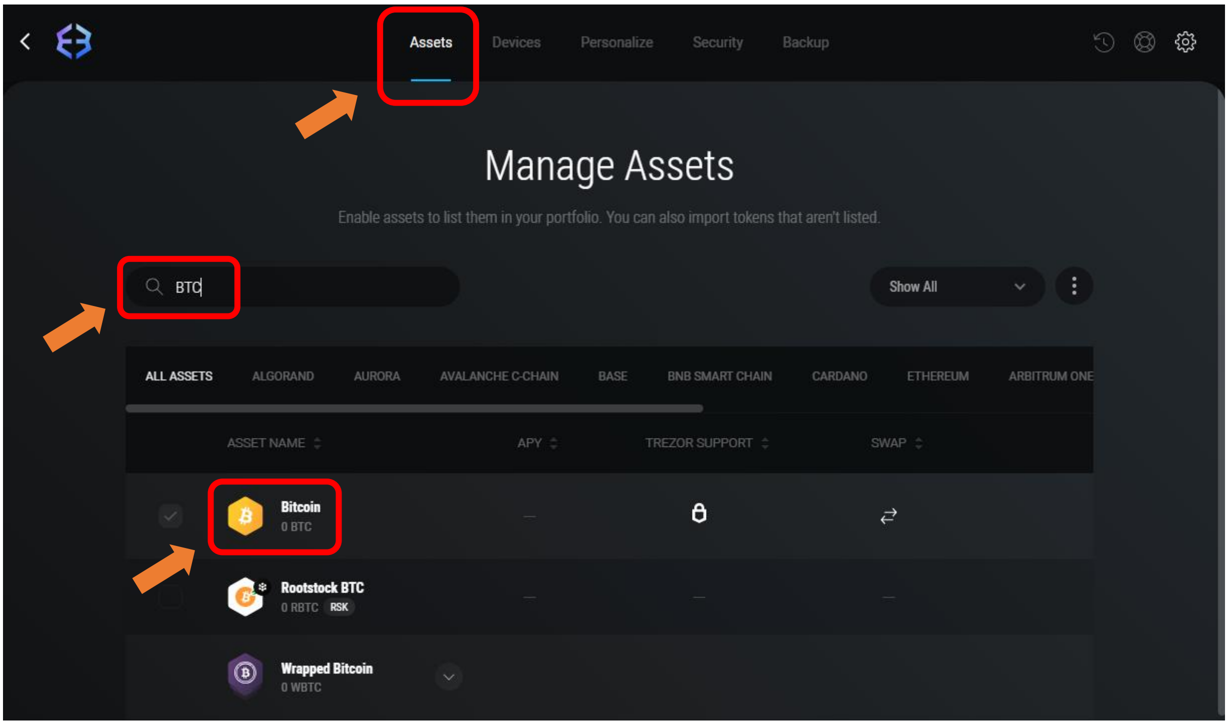
9. Check “**Show Secret Key**”.
   Please note: This interface cannot be screenshot or photographed. Do not disclose any information related to these mnemonic words to anyone. You must securely store these 12 mnemonic words corresponding to the numbers. Without them, you will permanently lose access to this wallet.
10. Click “**DONE**”.
11. Select the correct answer based on the recorded mnemonic words. Then click “**DONE**”.



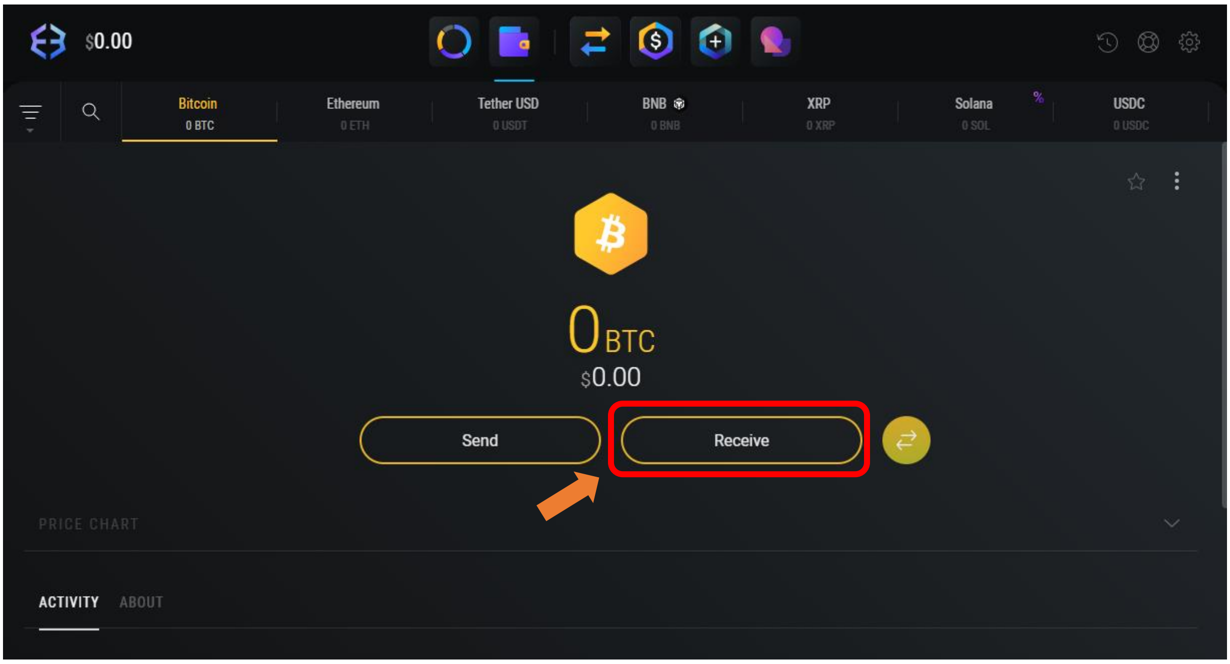
12. So far, you have successfully created your wallet.



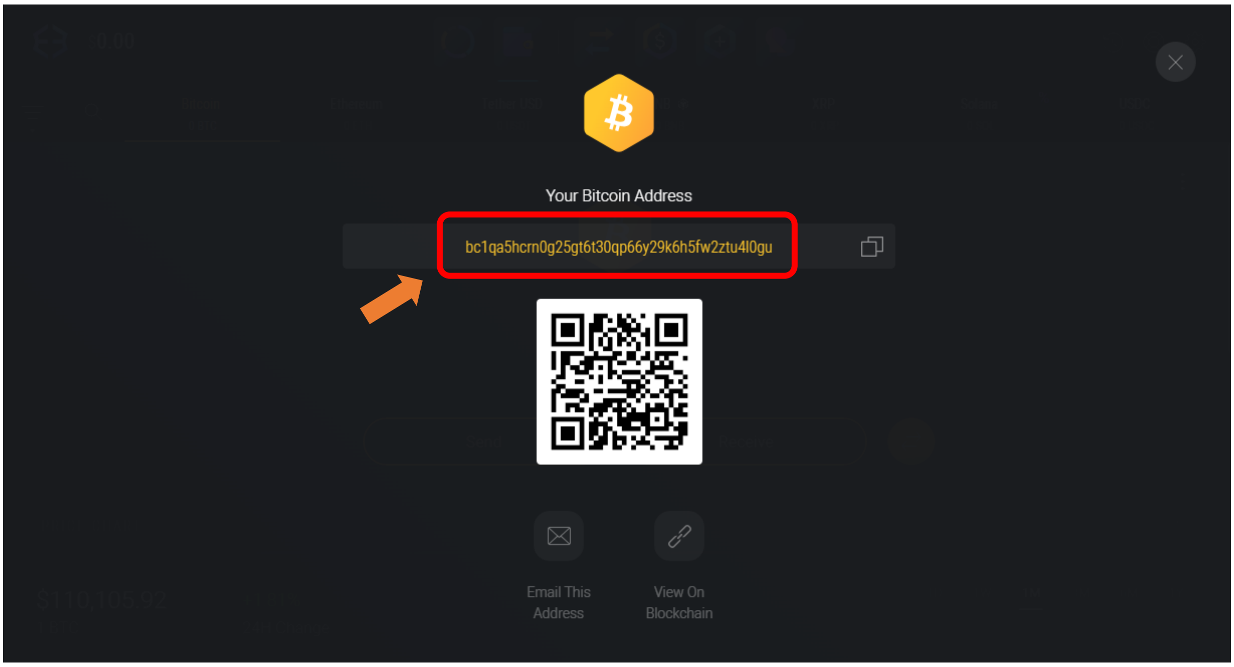
13. Click "**Assets**", type "**BTC**" to search, and then click "**Bitcoin**".



14. Click “**Receive**”.



15. The generated string of characters is a BTC address. This address can be publicly shared and is used to receive BTC.



For any Exodus-related usage questions, answers can be found at:
https://www.exodus.com/support/en/

If you have any concerns, please feel free to contact us via support@freenove.com

NerdMiner_v2 (Freenove Miner)

This project utilizes Freenove ESP32 Display and NerdMiner_v2 to implement Bitcoin mining. It requires some programming background and a basic understanding of digital currencies.

Project Introduction

NerdMiner_v2 (https://github.com/BitMaker-hub/NerdMiner_v2)is an excellent open-source BTC mining firmware designed for ESP32 development boards. Its goal is to allow users to experience the process of "mining a Bitcoin block" using a small piece of hardware. The primary aim of this project is to help more people **learn about cryptocurrency mining** while owning an aesthetically pleasing desktop accessory.

NerdMiner_v2 supports mining on **solo pools** by implementing the **Stratum protocol** (by default, it connects to **public-pool.io**, which supports Nerdminer devices, but the pool address can be customized). Freenove has ported this project to run on the Freenove ESP32 Display.

This tutorial will guide you through the process of running the project on the Freenove ESP32 Display.

Precautions

- **Cost**
  - As of the time of writing this document, NerdMiner_v2 is temporarily free to use.
- **Security Precautions**
  - Never disclose your wallet's private key or recovery phrase to anyone. The only information you need to provide is your public cryptocurrency receiving address.
- **Troubleshooting**
  - If you have followed the tutorial but are still unable to use the miner, please feel free to contact us for assistance. (**support@freenove.com**)

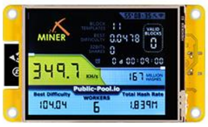
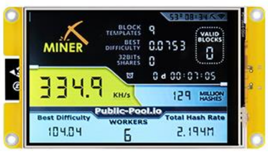
Compatibility Description

At the time of writing this guide, the Freenove ESP32 Display is available in five different models. Although they vary in terms of screen drivers, resolutions, and screen sizes, all of them can be set up using this tutorial to learn how to use NerdMiner_v2.

The official firmware of NerdMiner_v2 is not fully compatible with this product. Therefore, Freenove has conducted secondary development based on it to ensure stable operation.

Miner Compatibility

Provide a tutorial to show how to install the following miner firmware.

Model	**NerdMiner** (Freenove version, free) (Pre-installed in the product)	**NMMiner** (2.99 USD per device) (Need to buy and install it yourself)
2.8 Inch ST7789 TN	 fully compatible	 fully compatible
2.8 Inch ILI9341 TN	 fully compatible	 fully compatible
3.2 Inch ST7789 IPS	 fully compatible	 fully compatible, light theme
3.5 Inch ST7796 TN	 fully compatible	 fully compatible
4.0 Inch ST7796 TN	 fully compatible	 fully compatible

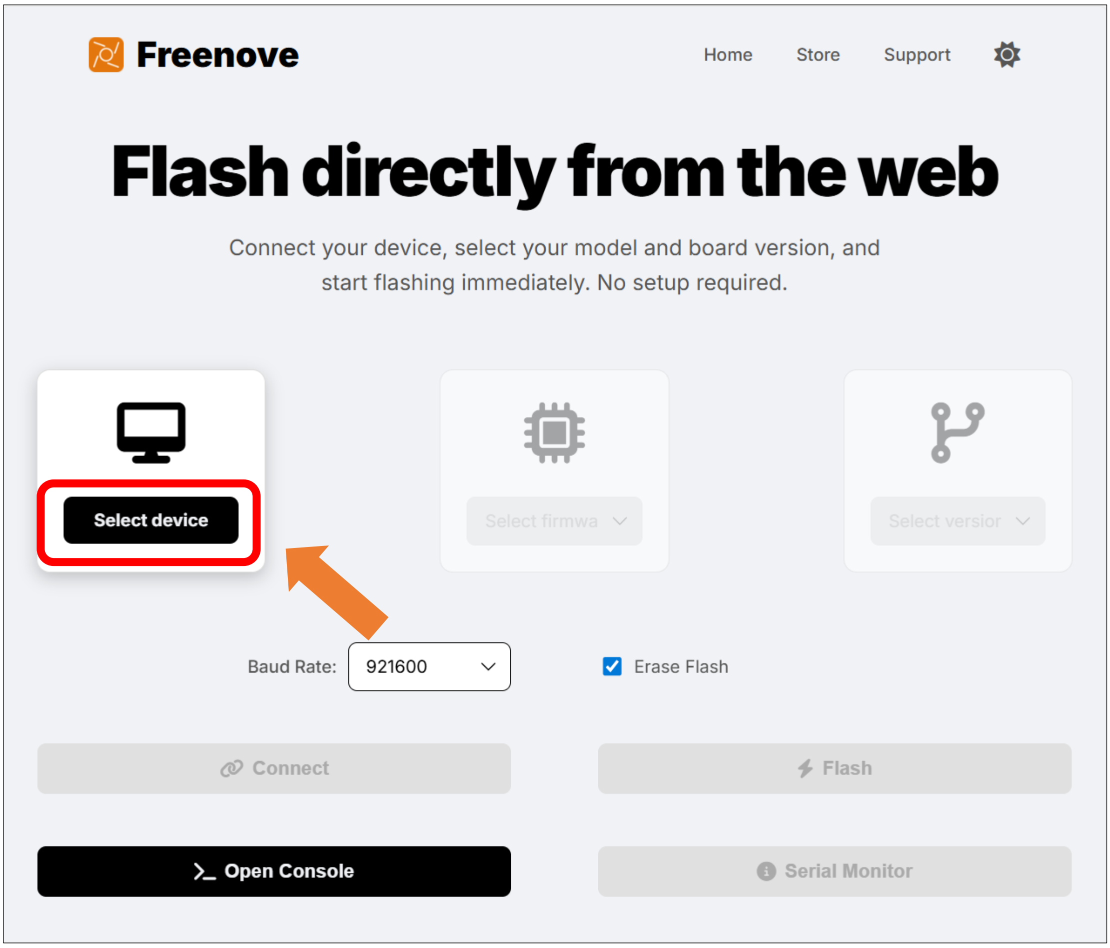
If you have any concerns, please feel free to contact us via support@freenove.com

How to Use

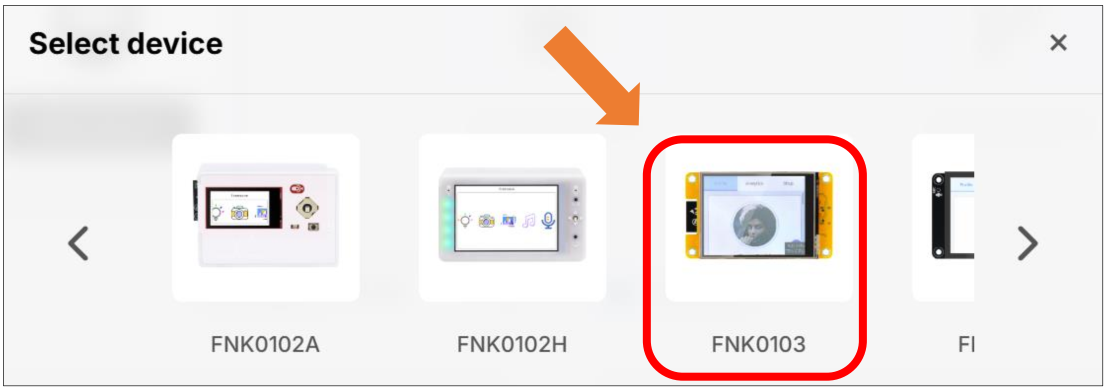
Firmware Flashing

Note: Since the official NerdMiner_v2 firmware is not compatible with this product, this tutorial only demonstrates how to flash the adapted firmware onto the product via Freenove official website. For information on the official NerdMiner_v2 firmware tool (Bitronics Flasher), please refer to the relevant documentation separately.

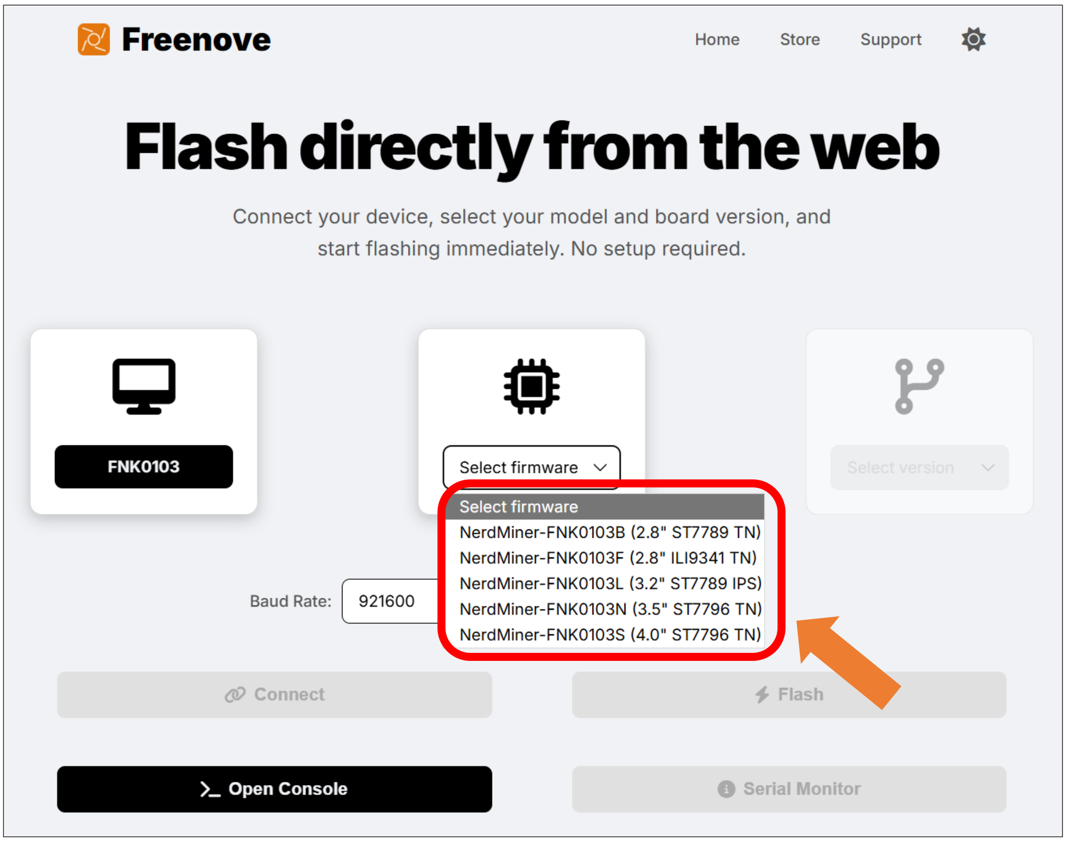
1. Open the Freenove - Web Flasher and click "**Select device**".



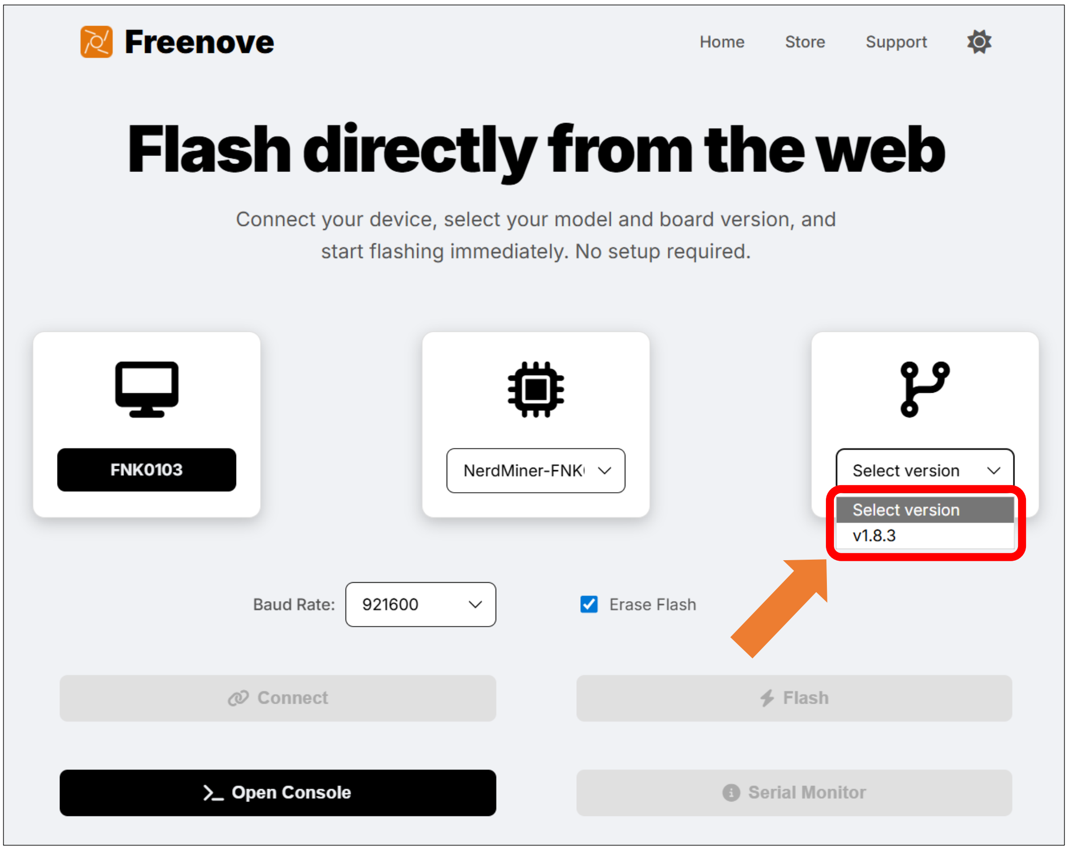
2. Select "**FNK0103**"



3. Click "**Select firmaware**" and select the firmware corresponding to your display model.



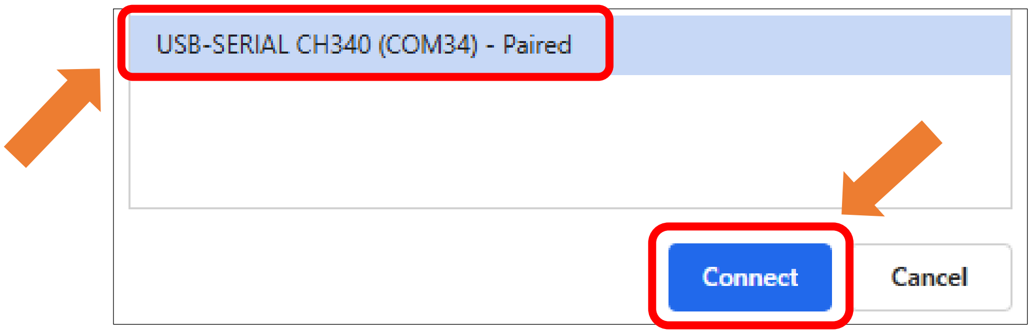
4. Click "**Select version**" to select the appropriate version.



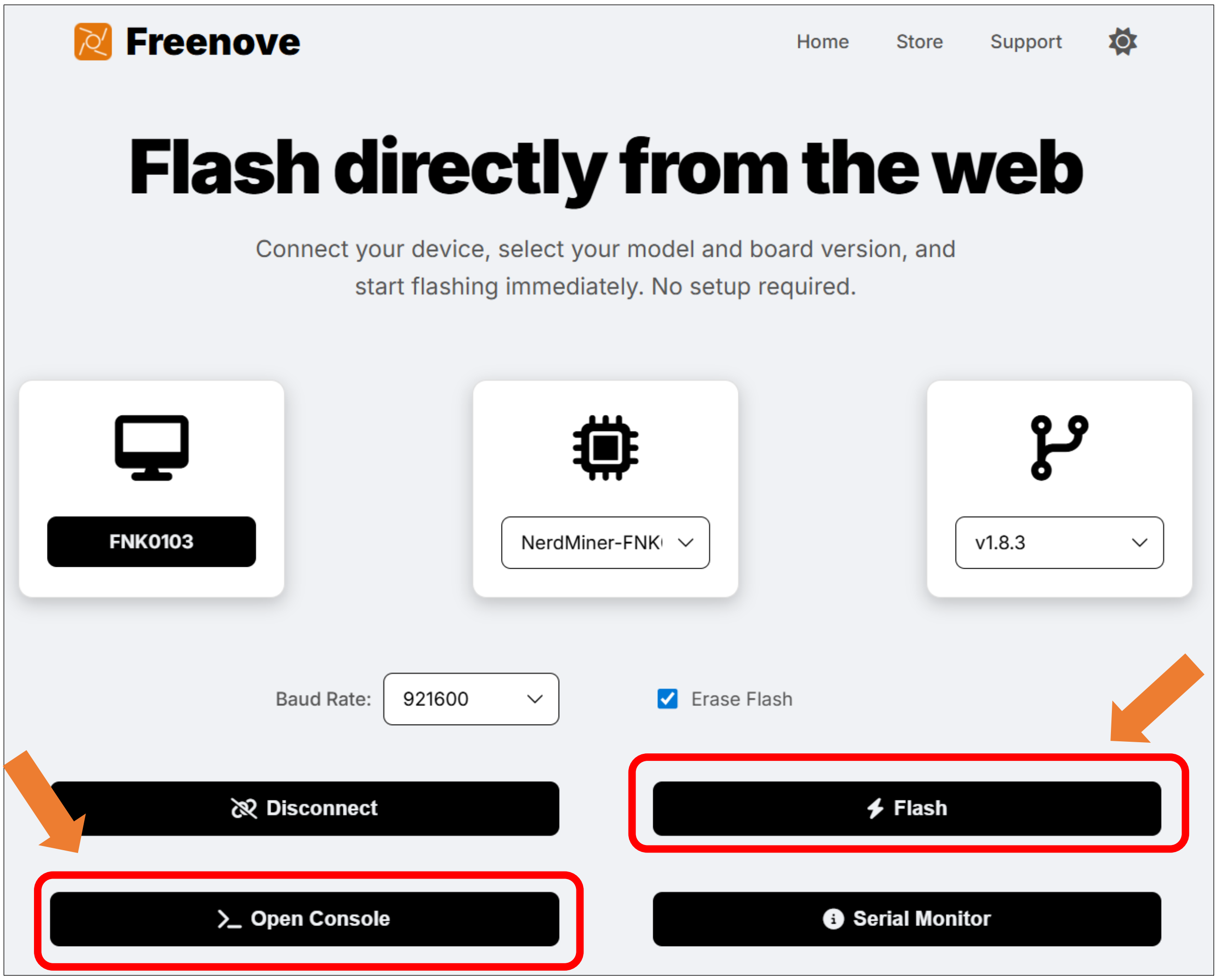
5. Click "**Connect**", select the correct serial port from the pop-up window in the upper left corner of the webpage, and then click "**Connect**" within that window.

Note:

1. **The port number (e.g., COMx) is dynamically assigned by your system. The specific value (such as COM3, COM5, etc.) may differ from the example in the diagram. Please select the port that is actually displayed.**
2. **COM1 is typically not the port for the target device. Please select a different port to connect.**



6. Click "**Flash**" to start the firmware burning process. Click "Open Console" to view the current progress.

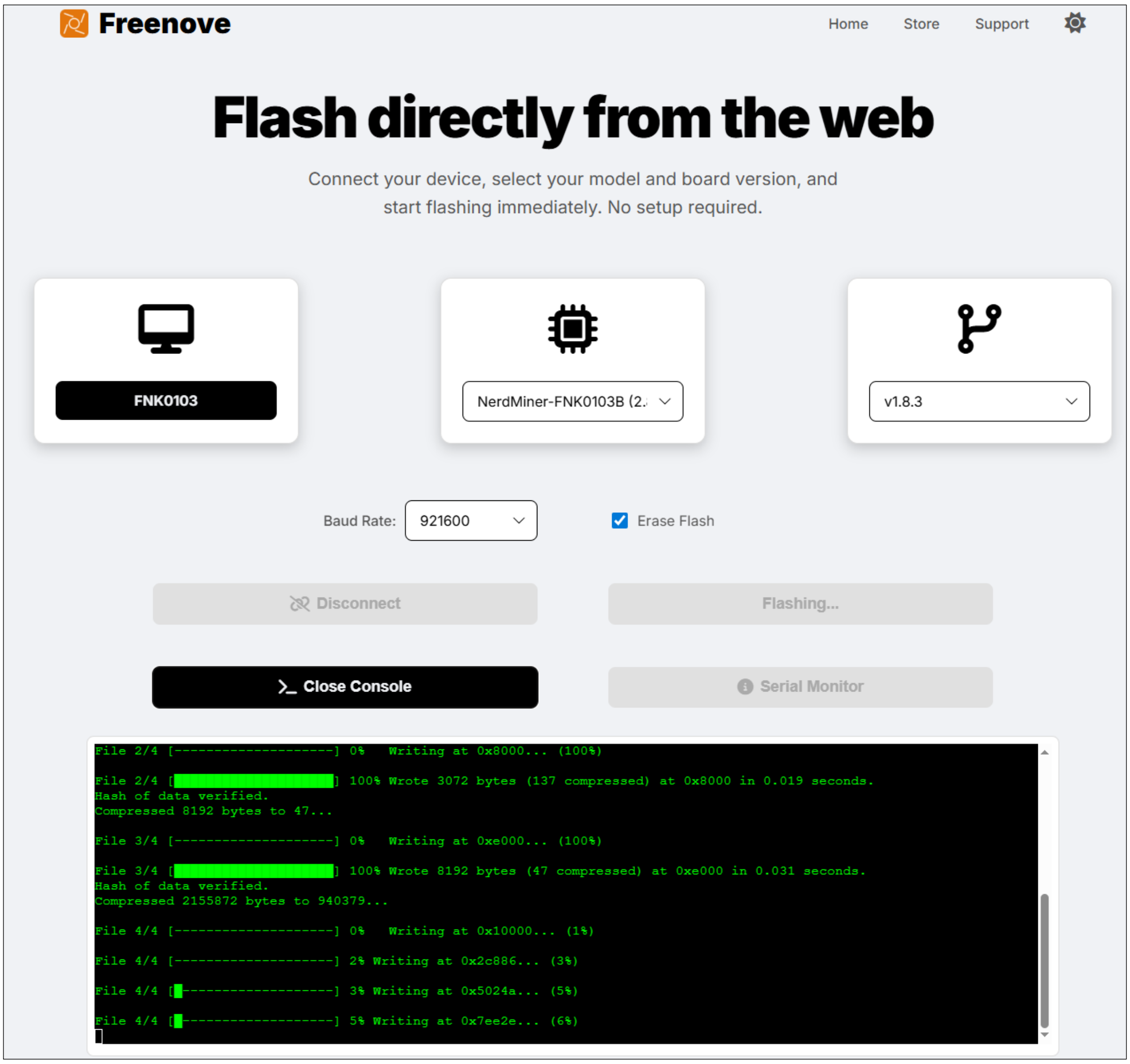


7. The device starts flashing the firmware.

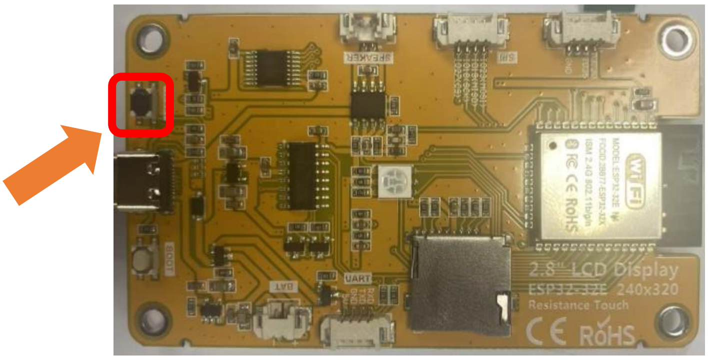
Note:

1. **Do not disconnect the device from your computer during the firmware flashing process!**
2. **If the download does not start after clicking "Flash", it may be due to network latency. Try refreshing the webpage and attempting the process again.**
3. **If the "Flash" button is disabled (grayed out), please check that all previous configuration steps have been completed.**

If you have any concerns, please feel free to contact us via support@freenove.com



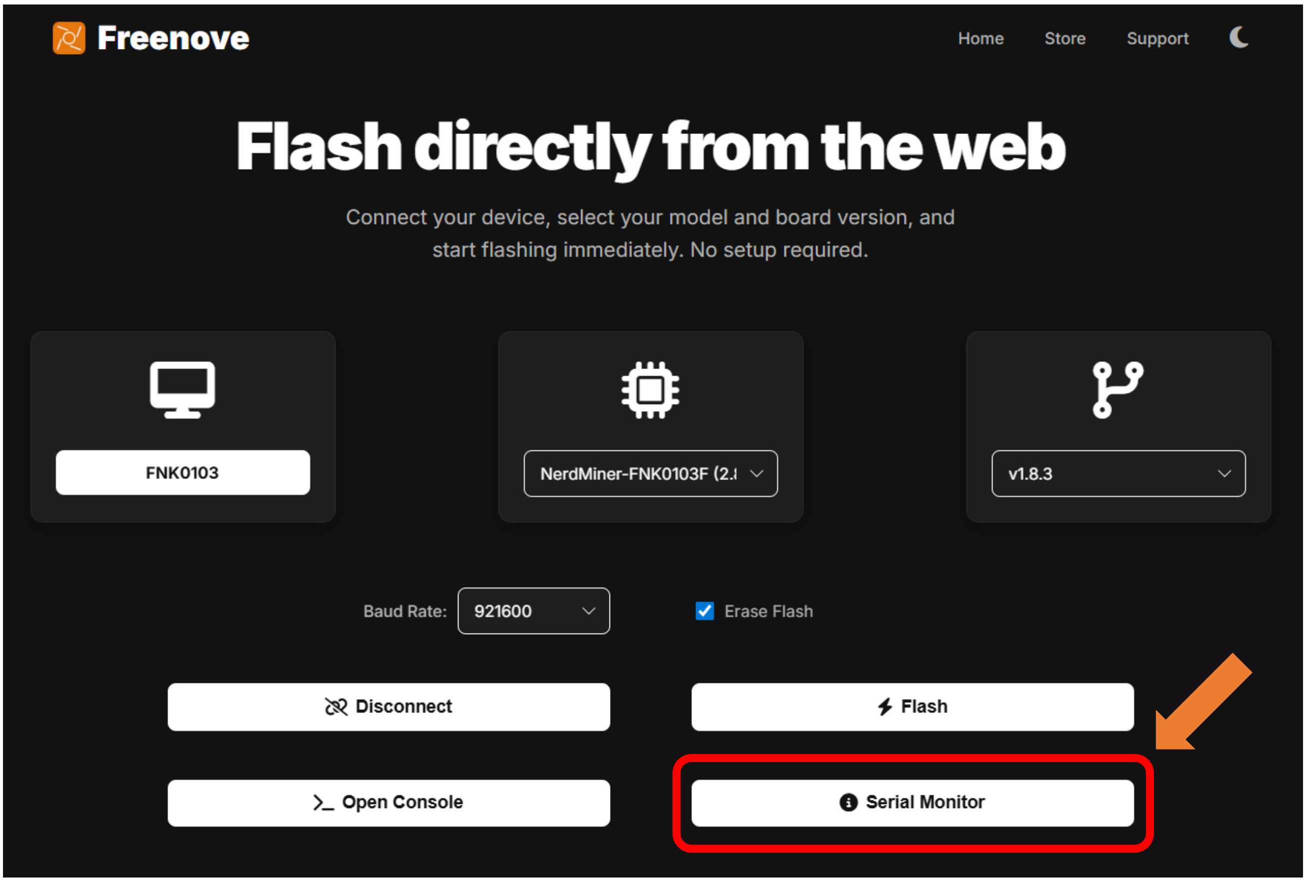
8. Click the reset button on the device.



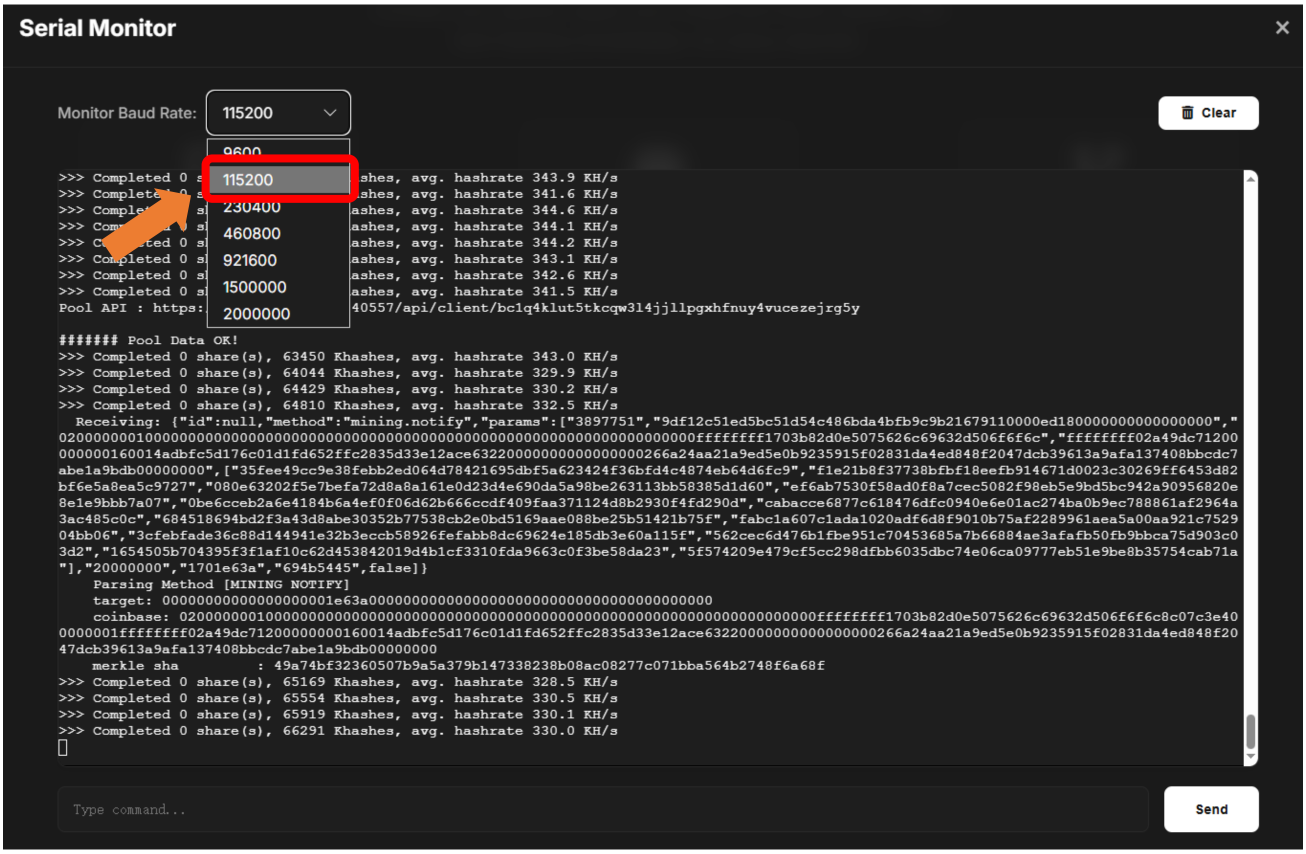
The device starts to work.

Note: During the first boot, the screen may display a solid color (e.g., white or other colors) for a few seconds. Please wait patiently, as this is normal and does not affect functionality.

9. (Optional) Click "**Serial Monitor**" to open the serial monitor.

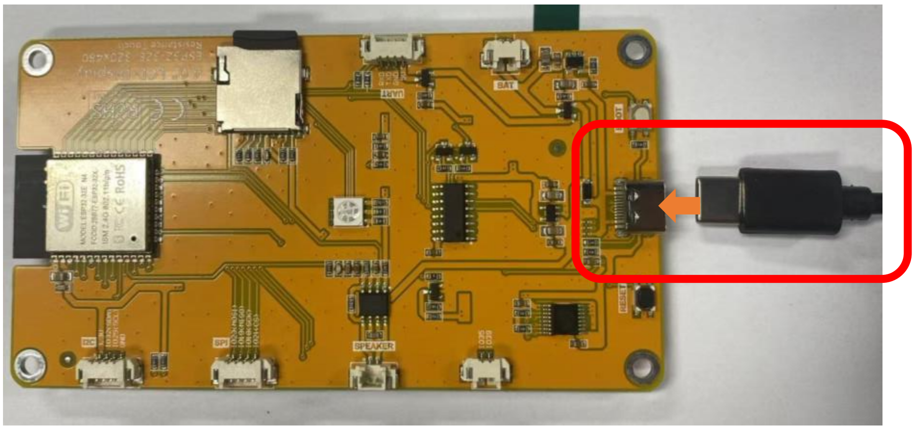


Set the baudrate to 115200, the serial monitor will output debugging information in real-time.



Wi-Fi Configuration

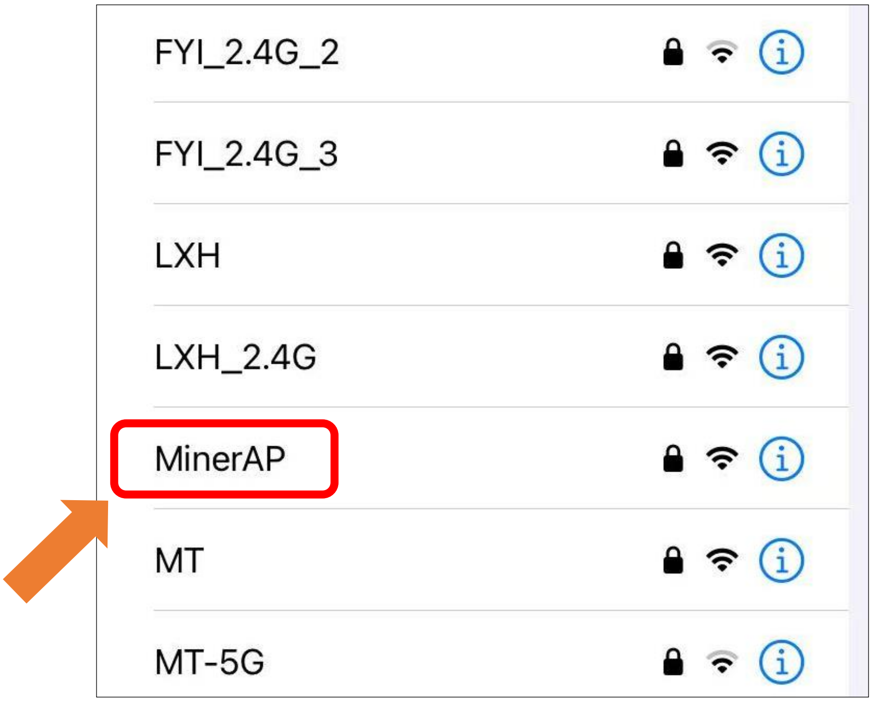
1. Connect the Freenove ESP32 Display to your computer. If the firmware has not been flashed yet, please navigate back to the Firmware Flashing Section to do it.



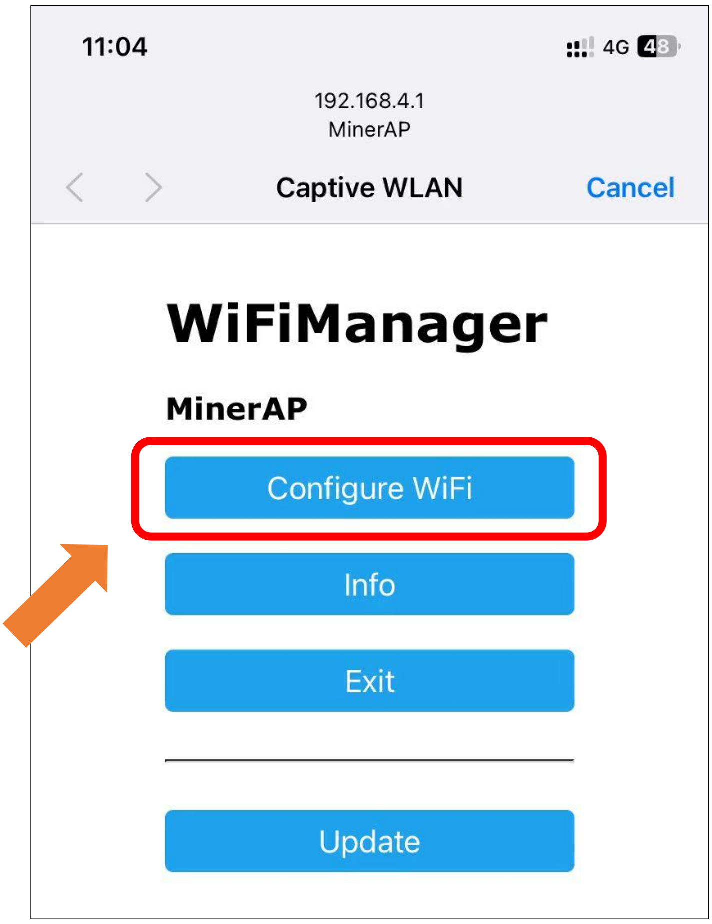
2. Turn ON WLAN and connect to **MinerAP**

WiFi SSID： MinerAP

PASSWORD： MineYourCoins

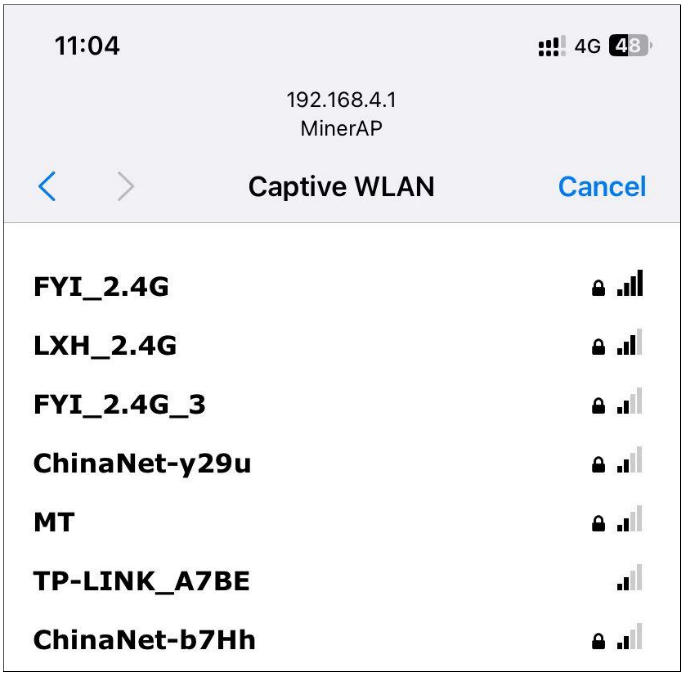


3. Click "**Configure WiFi**"

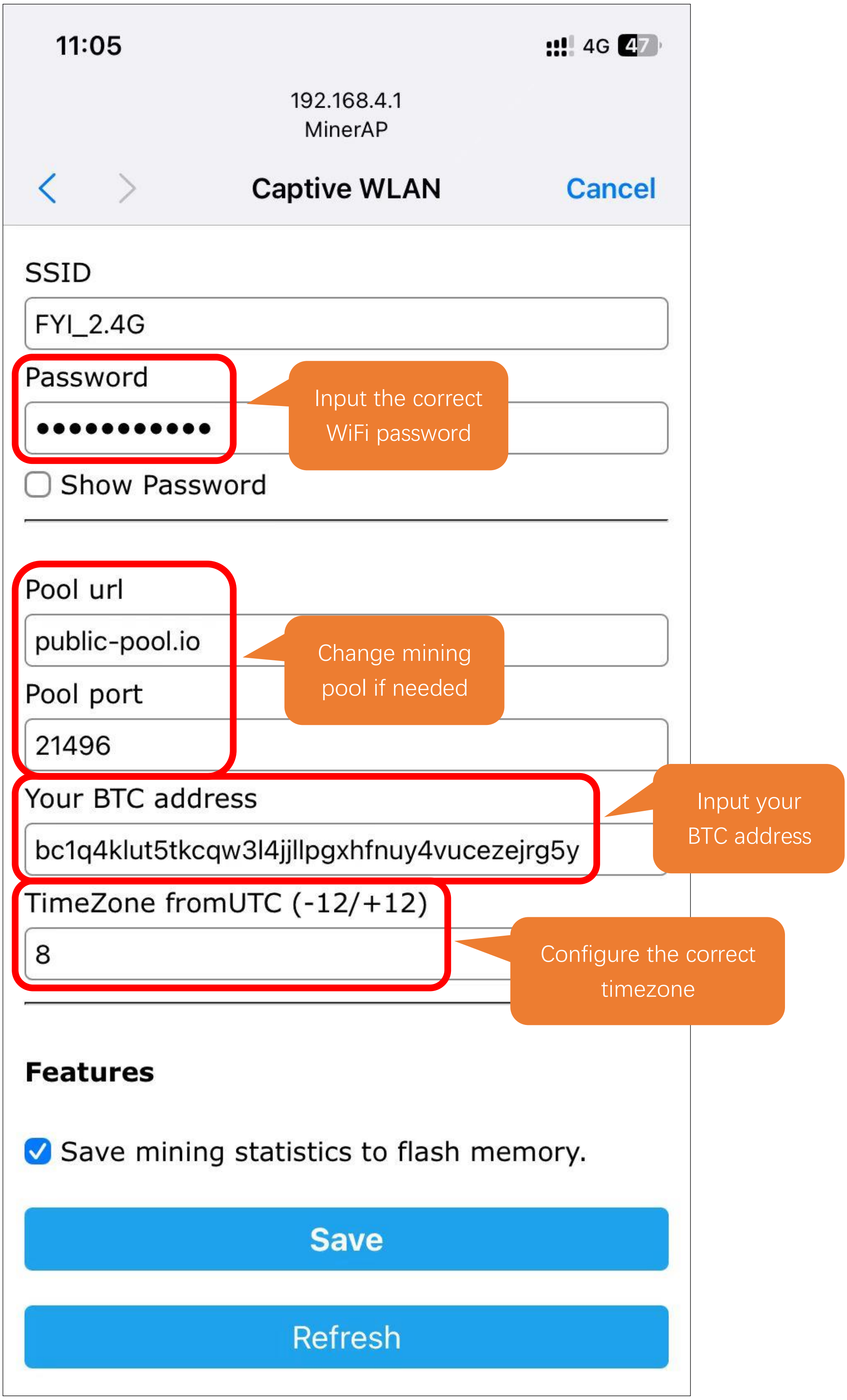


Note: If the page does not redirect automatically, please visit 192.168.4.1 in your browser.

4. Select your network. Please note that ESP32 **only supports 2.4GHz network.**



5. Configure device information.
6. If you do not have a BTC address yet, please refer to Obtaining the BTC Receiving Address Section to get it.
7. **If the mining efficiency is unsatisfactory, you can switch to other mining pools as needed.**

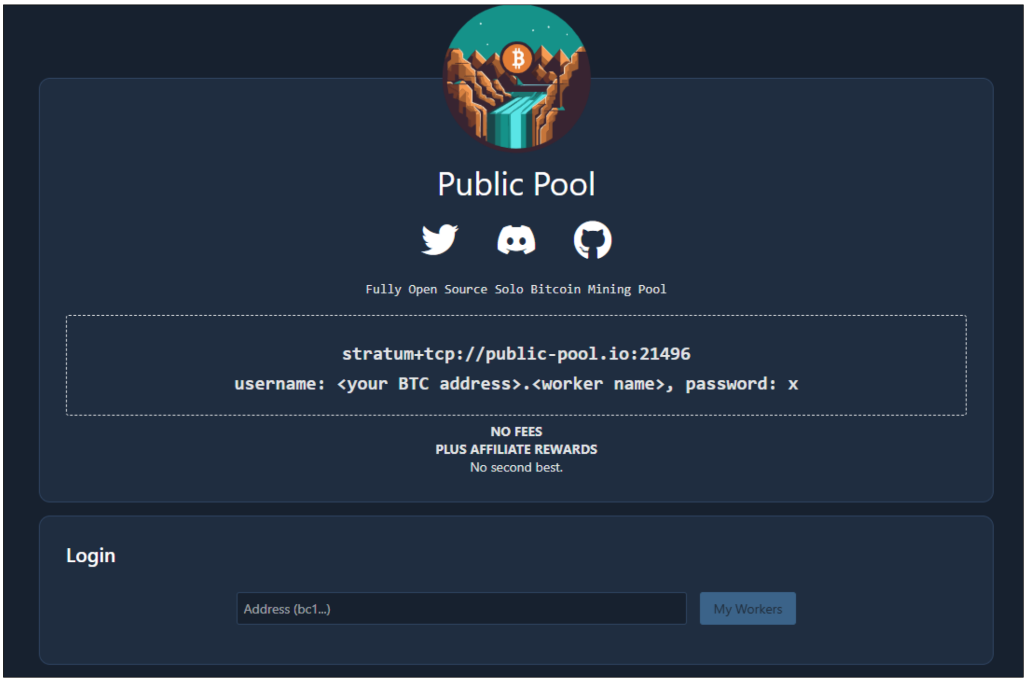


Management Tool

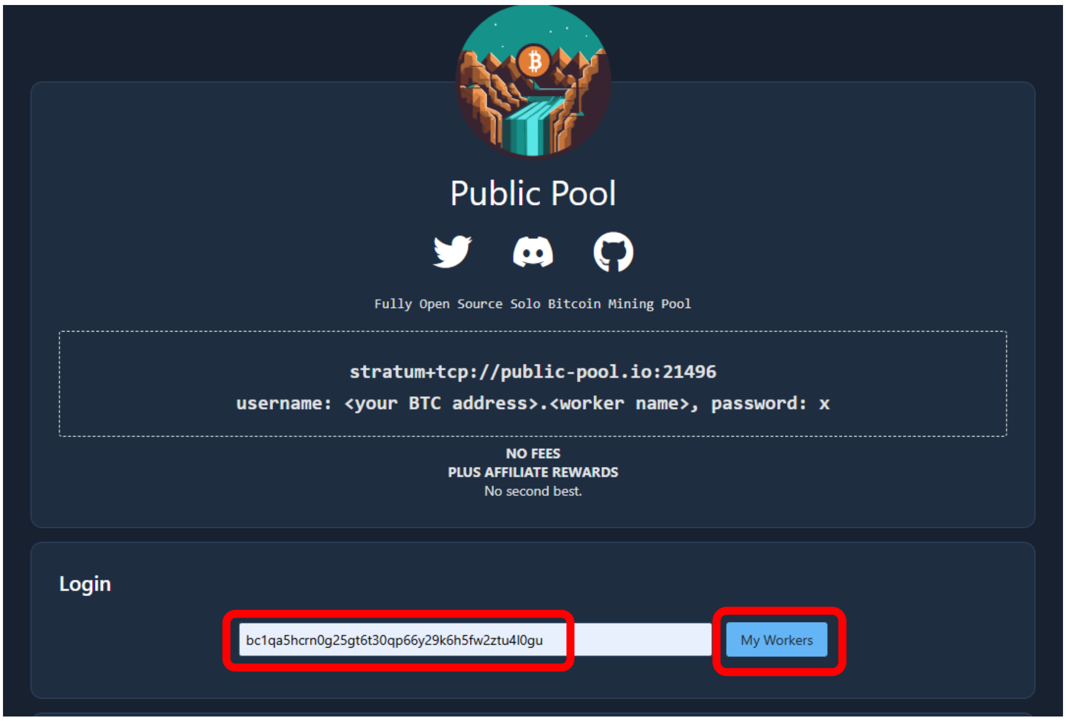
Public Pool

When using the pool public-pool.io:21496, you can manage it via the web link https://web.public-pool.io/#/.

1. Open public-pool



2. Enter your BTC receiving address, and click "**My Workers**".



3. You can view the current working status of the worker.

Please note: This webpage only detects and manages miners currently working in the Public Pool that use your BTC address as the login or payout address. Once the pool is changed, this webpage will no longer be accessible for management.

NMMiner

This project utilizes Freenove ESP32 Display and NMMiner to implement Bitcoin mining. It requires some programming background and a basic understanding of digital currencies.

Project Introduction

NMMiner (https://github.com/NMminer1024/NMMiner) is an excellent BTC mining firmware specifically designed for ESP32 development boards. It breaks away from the traditional mining requirements of massive electricity consumption and high hash rates by adopting a SOLO mining mode. In this mode, every computation has the chance to independently win the entire block reward. NMMiner supports all cryptocurrencies based on the SHA256d algorithm, such as **BTC, BCH, XEC, and DGB**. Freenove has successfully ported this project to run on the Freenove ESP32 Display.

This tutorial will guide you through the process of running the project on the Freenove ESP32 Display.

NMMiner offers two management tools:
1. **NMController Client**: Windows desktop application
2. **NMController Web**: A cross-platform web client designed with Python, compatible with both Windows and macOS systems.

Precautions

- **Cost**
    - NMMiner is currently a paid service. Each device requires a unique license for activation. Therefore, you will need to purchase a separate activation code for every development board you plan to use in order to complete the activation process.

- **Security Precautions**
    - Never disclose your wallet's private key or recovery phrase to anyone. The only information you need to provide is your public cryptocurrency receiving address.

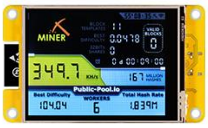
- **Troubleshooting**
    - If you have followed the tutorial but are still unable to use the miner, please feel free to contact us for assistance. (**support@freenove.com**)

Compatibility Description

At the time of writing this guide, the Freenove ESP32 Display is available in five different models. Although they vary in terms of screen drivers, resolutions, and screen sizes, all of them can be set up using this tutorial to learn how to use NMMiner.

Miner Compatibility

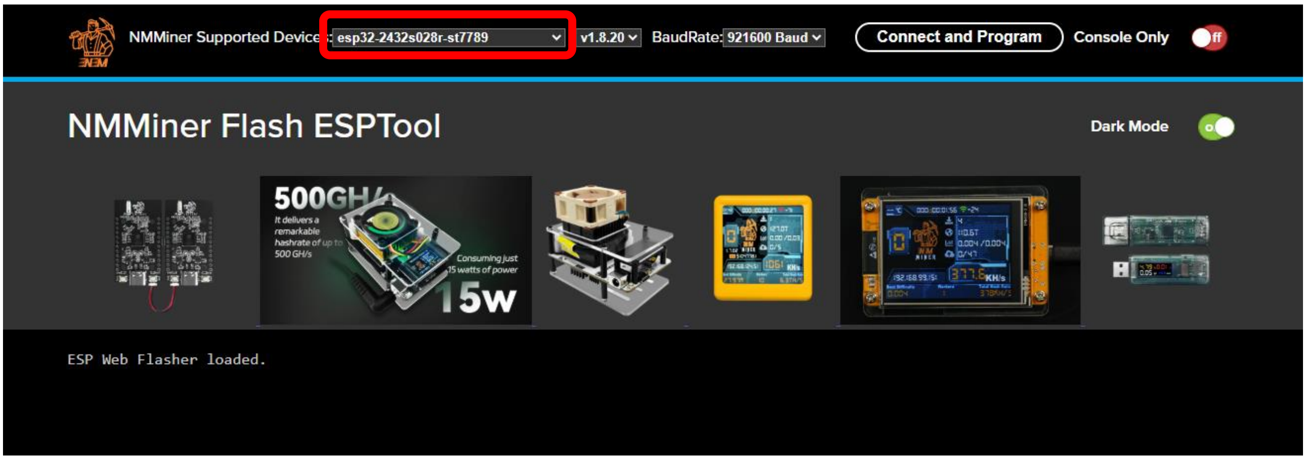
Provide a tutorial to show how to install the following miner firmware.

Model	**NerdMiner** (Freenove version, free) (Pre-installed in the product)	**NMMiner** (2.99 USD per device) (Need to buy and install it yourself)
2.8 Inch ST7789 TN	 fully compatible	 fully compatible
2.8 Inch ILI9341 TN	 fully compatible	 fully compatible
3.2 Inch ST7789 IPS	 fully compatible	 fully compatible, light theme
3.5 Inch ST7796 TN	 fully compatible	 fully compatible
4.0 Inch ST7796 TN	 fully compatible	 fully compatible

How to Use

Firmware Flashing and Activating

1. Open the NMMiner Flash ESPTool interface, click to select the device.

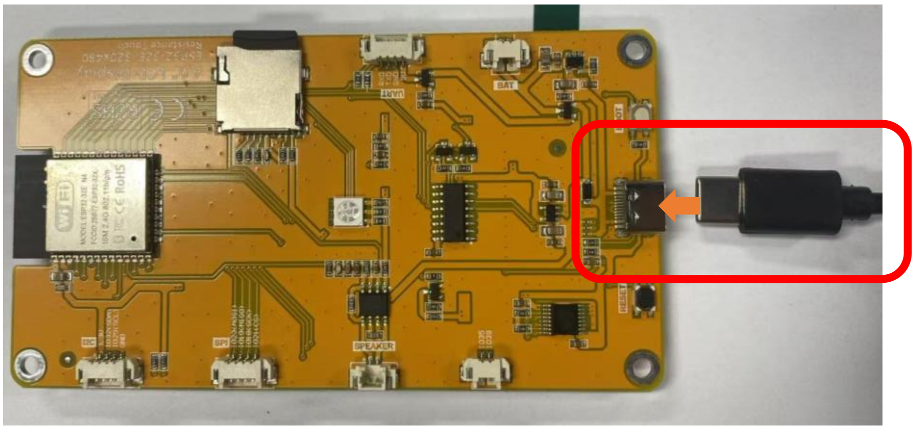


2. Select the device corresponding to your display according to the table below.

Models	Device Name
FNK0103F_2P8	esp32-2432s028r-ili9341
FNK0103B_2P8	esp32-2432s028r-st7789
FNK0103L_3P2	esp32-2432s024
FNK0103N_3P5	esp32-3248s035
FNK0103S_4P0	esp32-3248s035

If you have any concerns, please feel free to contact us via support@freenove.com

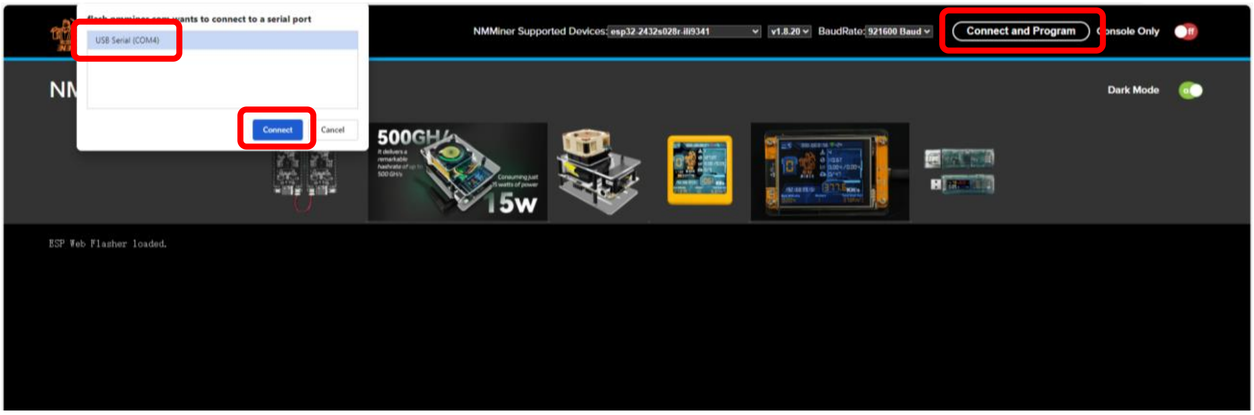
3. Connect the Freenove ESP32 Display to your computer.



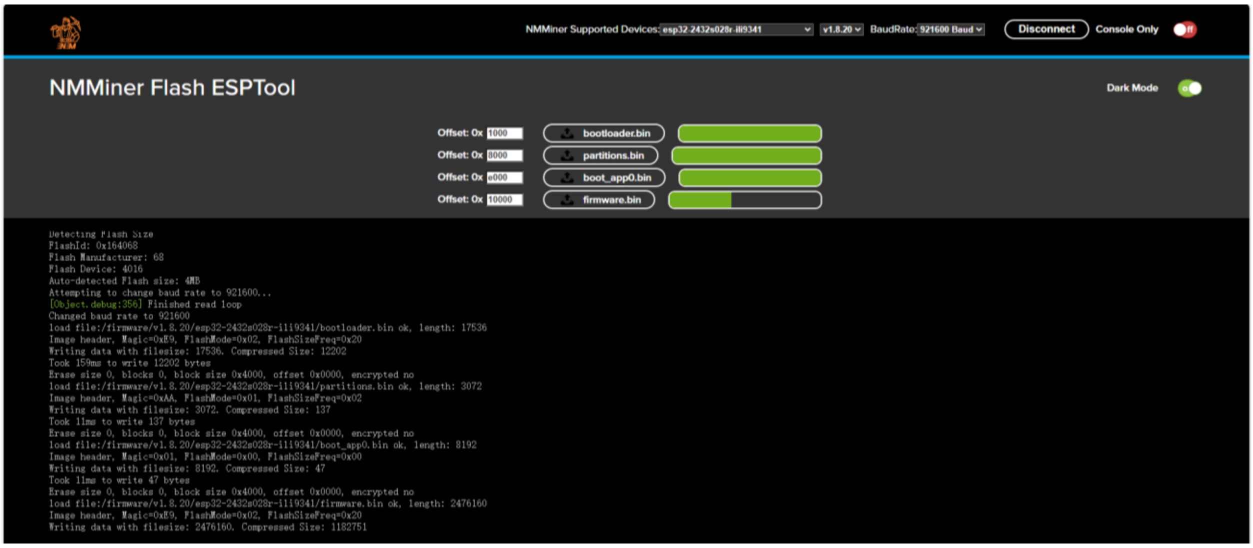
4. Click "**Connect and Program**", select the correct port in the pop-up window, and click "**Connect**".

Please note:

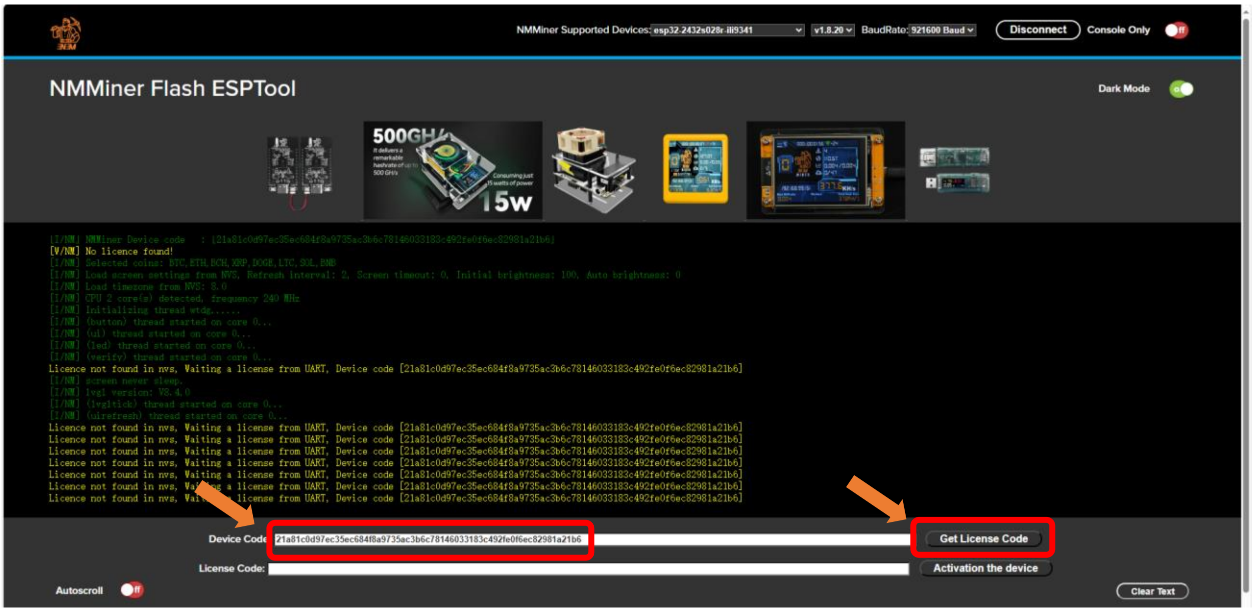
- The port may vary among computers
- Generally, COM1 is not the serial port of the ESP32.



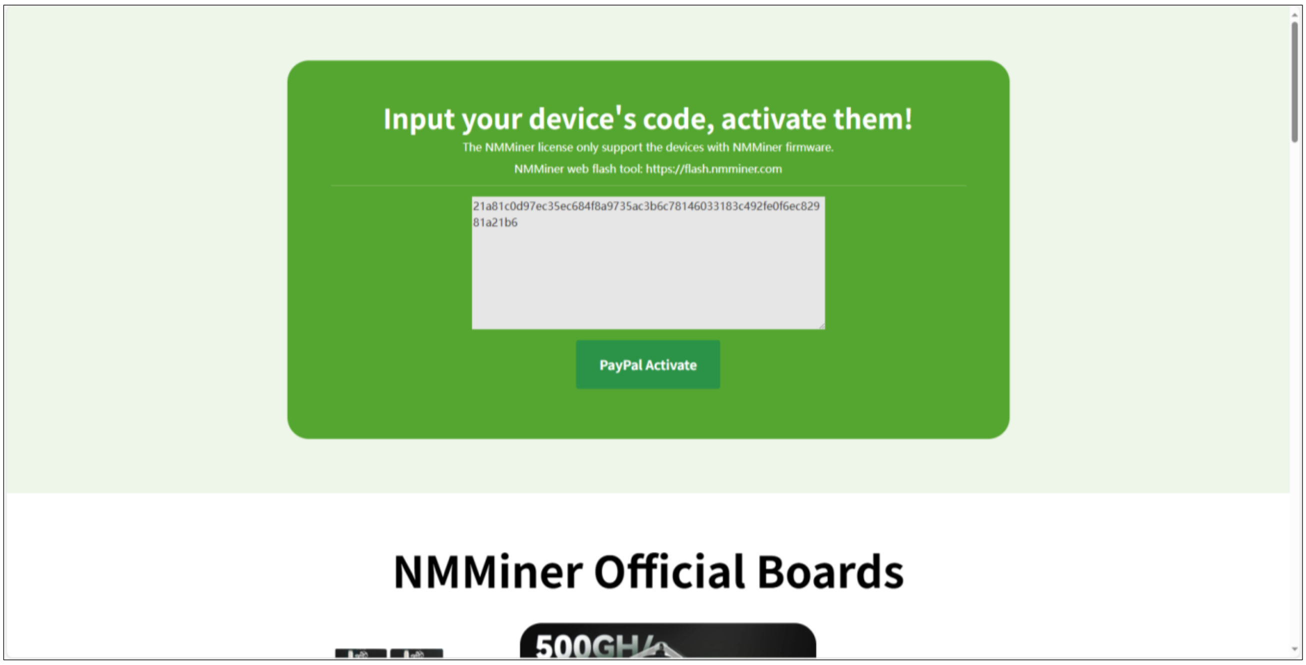
5. The firmware starts to flash into the Freenove ESP32 Display. Do NOT remove power supply during this process.



6. Click "**Get License Code**" after the firmware successfully flashes.

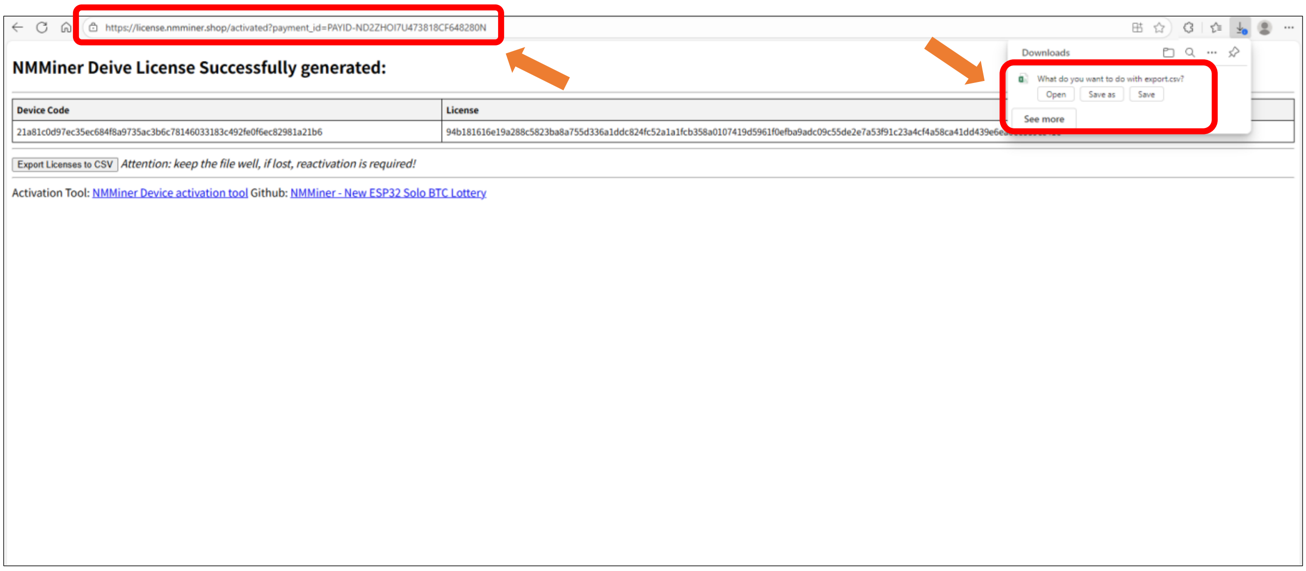


7. Click "**PayPal Activate**" to make payment.



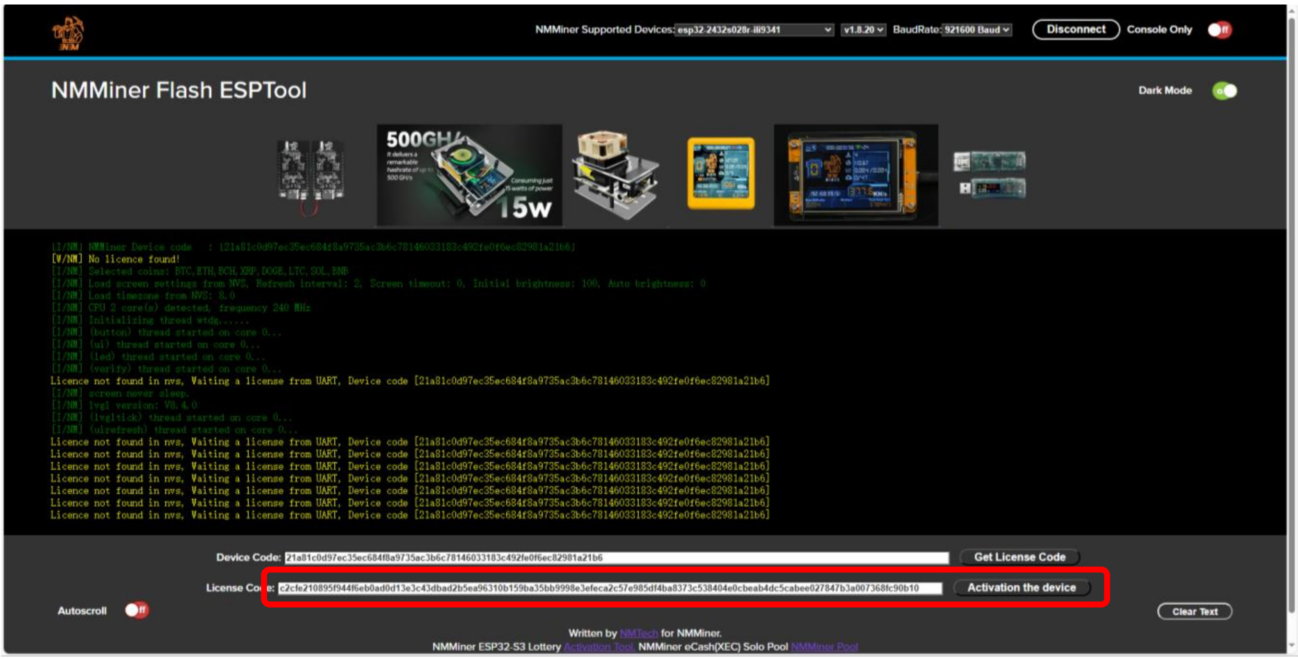
8. Please securely store the generated License (important). **If the License is lost, it will need to be repurchased.**

   When generating the license, the website will automatically download a CSV file containing the License information. You may also save the current webpage URL, as it can be reopened later.

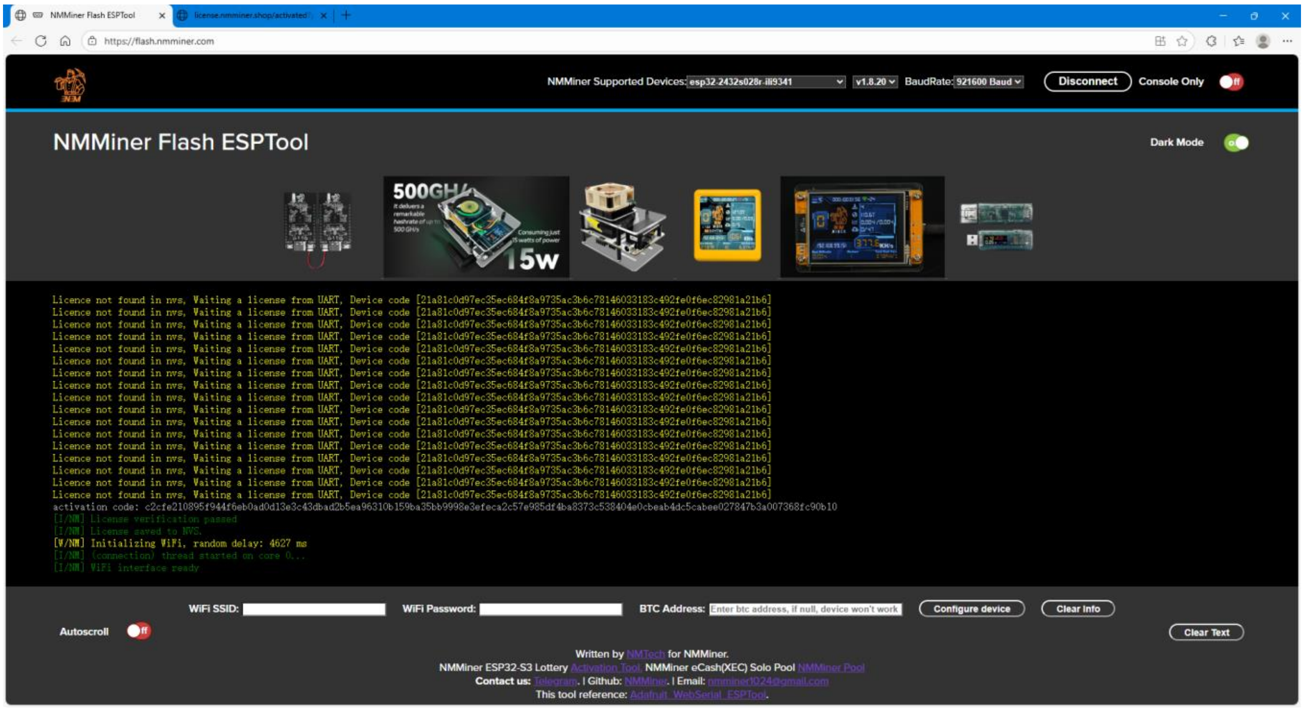


Please note: Each device requires a unique license code for activation. Therefore, you will need to purchase a separate activation code for every development board you plan to use in order to complete the activation process.

9. Enter the License Code and click “Activation the device”.



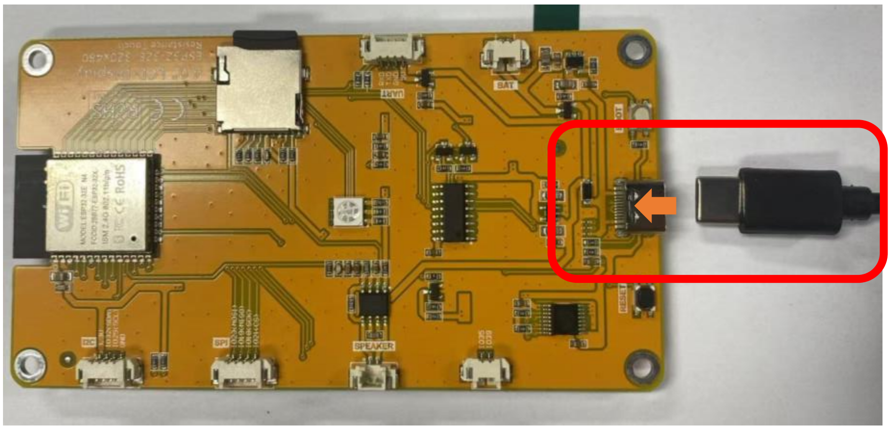
10. Now the device is successfully activated.



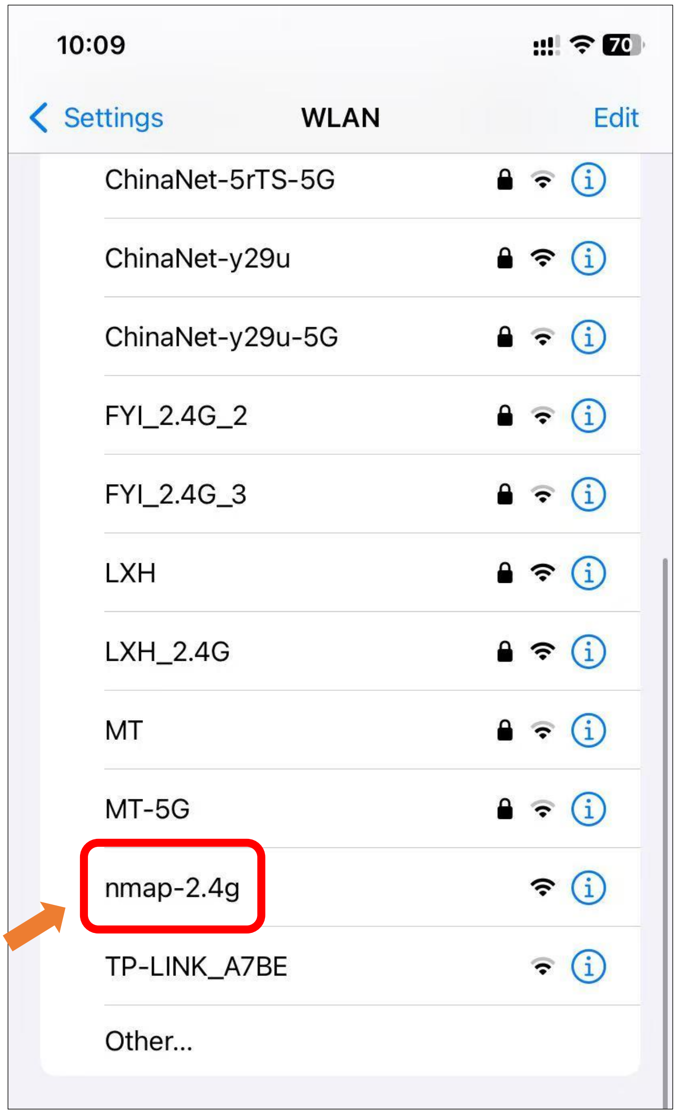
If you have any concerns, please feel free to contact us via support@freenove.com

Wi-Fi Configuration

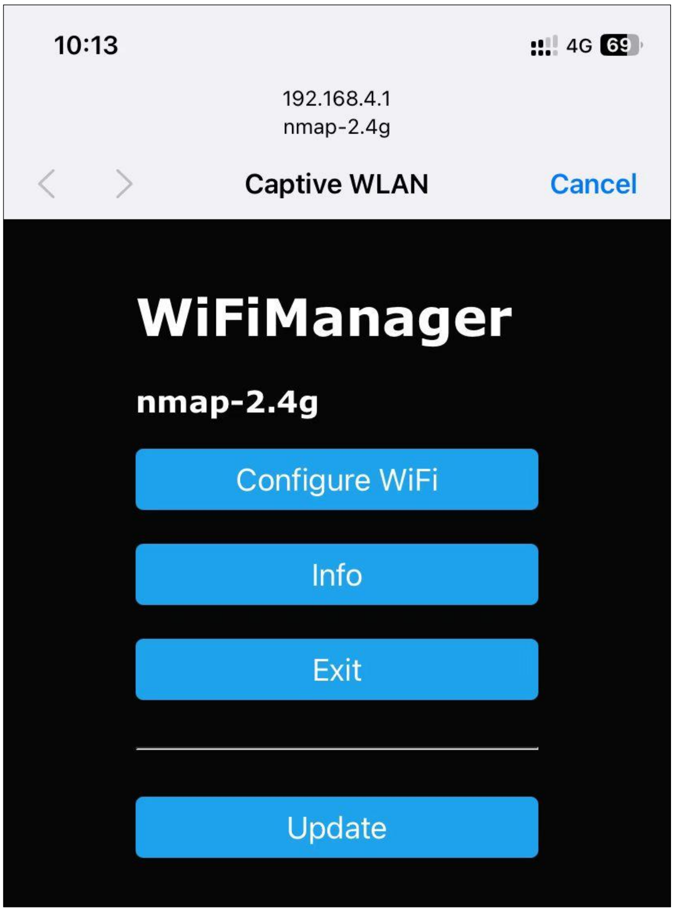
1. Connect the Freenove ESP32 Display to your computer. If the firmware has not been flashed yet, please navigate back to the previous section to do it.



2. Turn ON WLAN, locate and click "nmap-2.4g".



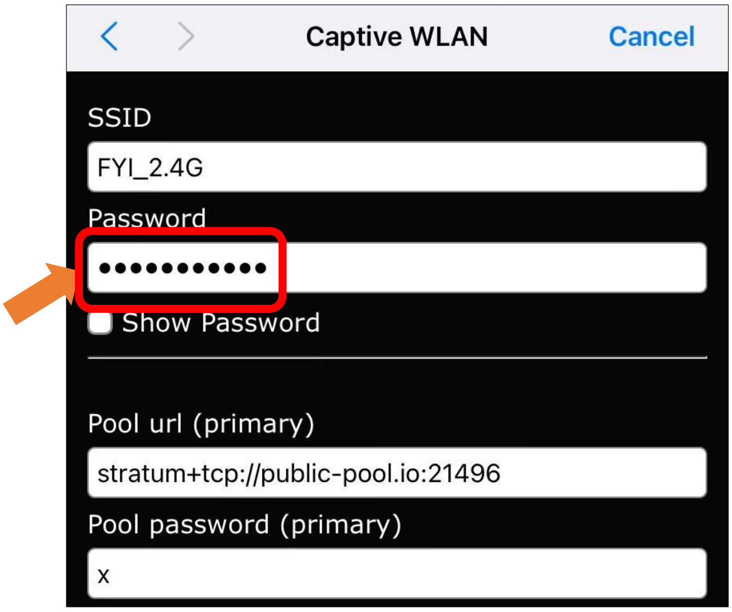
3. Wait for it to pop up the following interface, click "**Configure WiFi**".



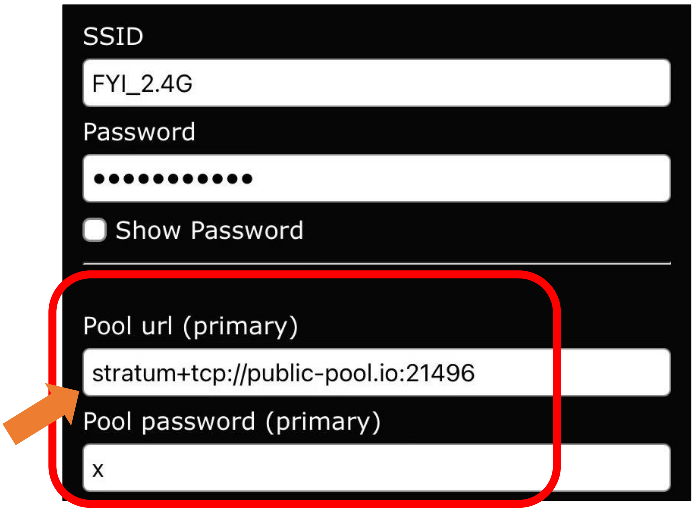
Note: If the page does not redirect automatically, please visit 192.168.4.1 in your browser.

4. Select your network. Please note that ESP32 **only supports 2.4GHz network**.

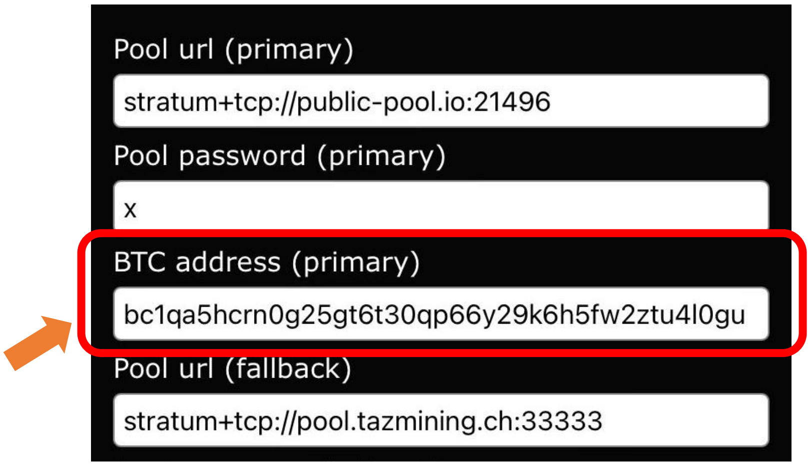
5. Enter the correct WiFi password.



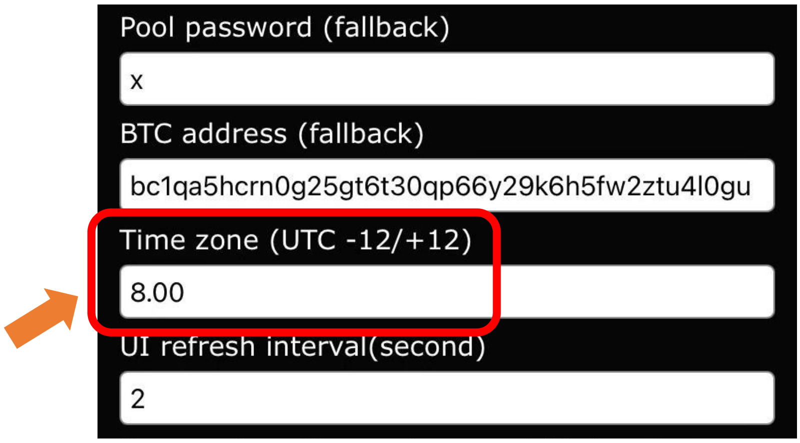
6. **Enter the mining pool information. If the mining efficiency is unsatisfactory, you can switch to other mining pools as needed.**



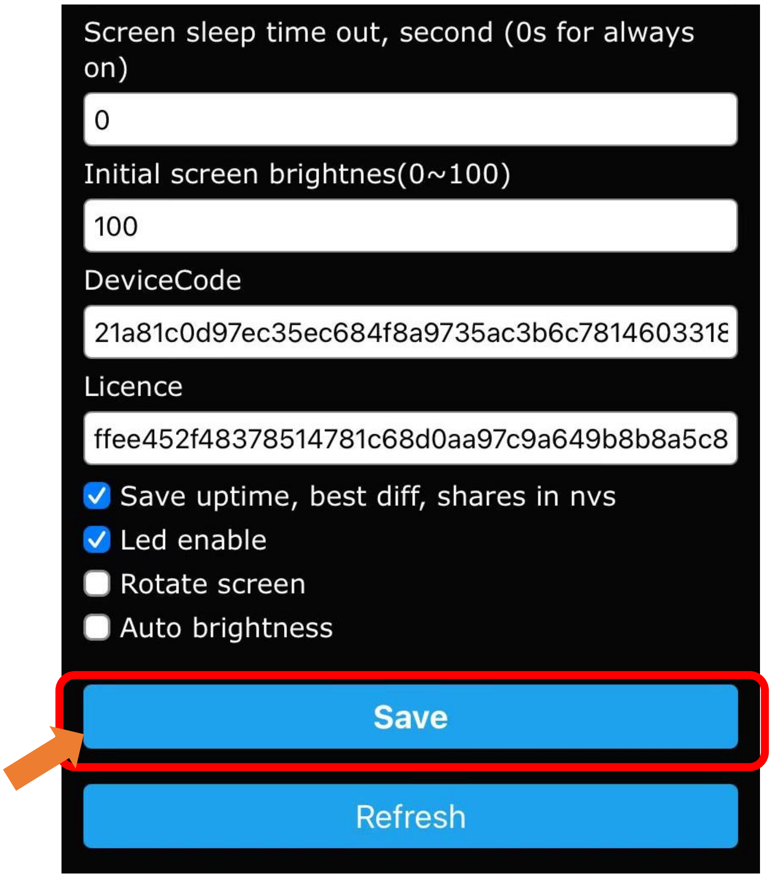
7. Enter the correct BTC address. If you do not have a BTC address yet, please refer to Obtaining the BTC Receiving Address Section to get it.



8. Enter the correct time zone.



9. Click “**Save**” to save the configuration.



10. Now the NMMiner project begins to work.

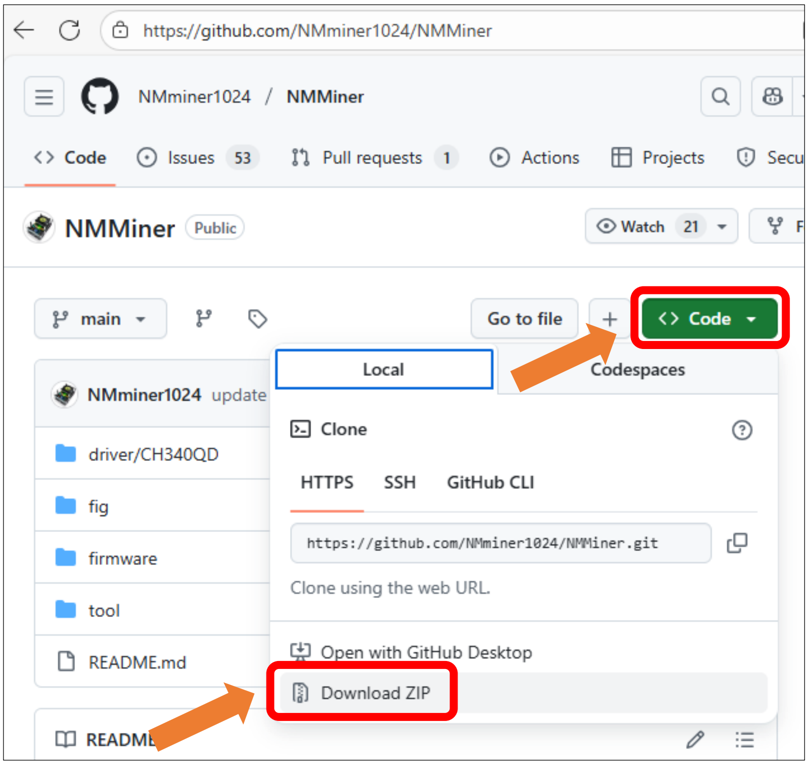
Management Tools

NMController Client

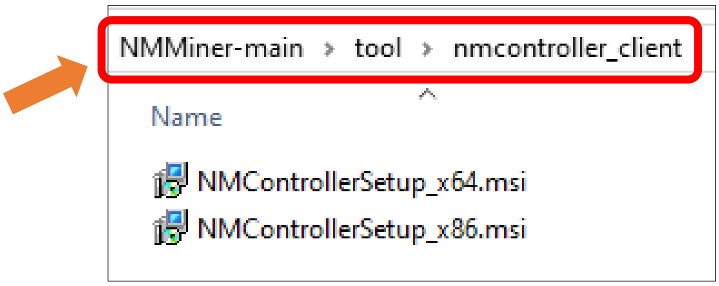
Note:

1) **NMController Client only works on Windows.**
2) **The NMController client, which is downloaded from NMMiner official, may trigger false positives from antivirus programs, incorrectly flagging it as a virus. If you do not trust it, you can use alternative methods provided in this document to manage your device.**

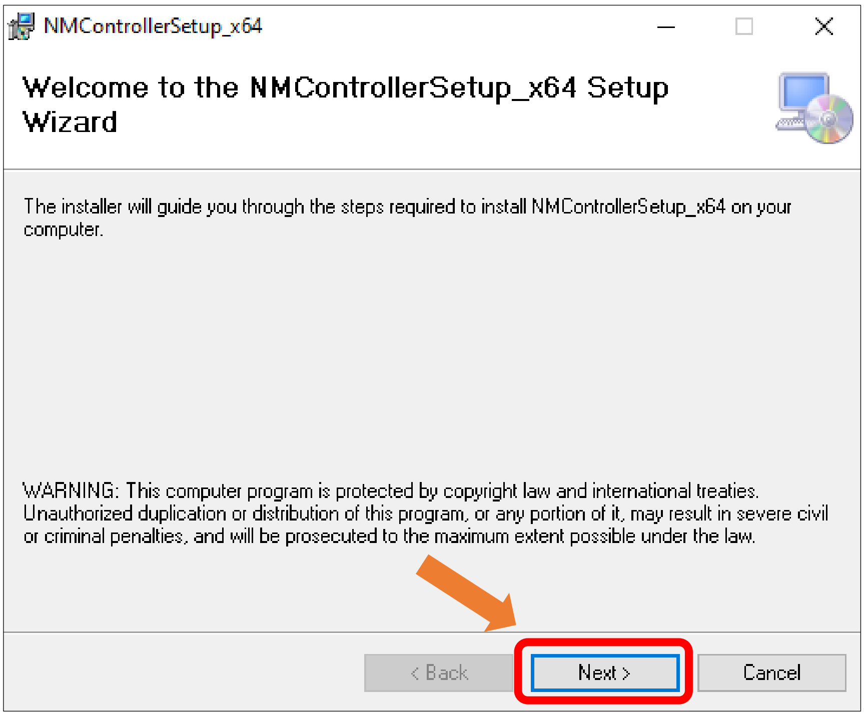
1. Click to open the [NMMiner GitHub website], click **Code** -> **Download ZIP**



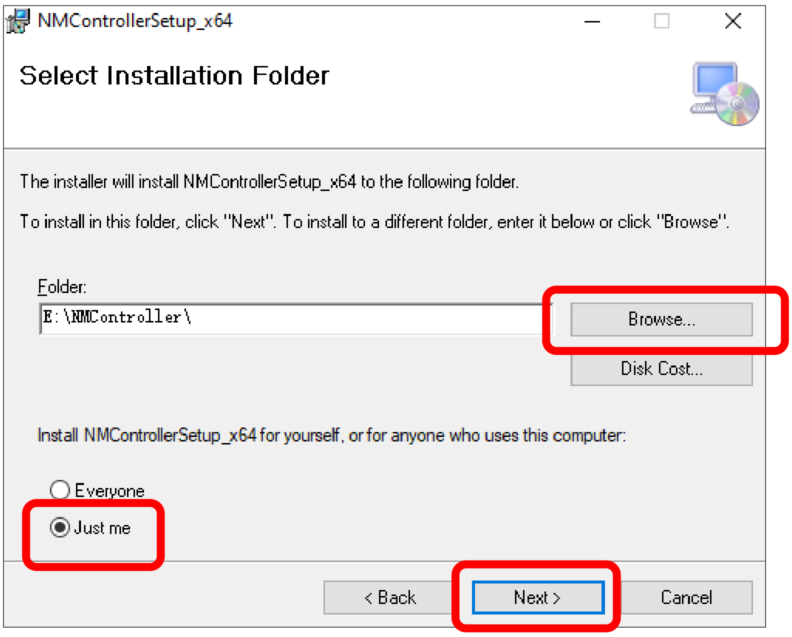
2. Extract the downloaded files, click **tool** -> **nmcontroller_client**, double click "**NMControllerSetup_x64.msi**" to open it.



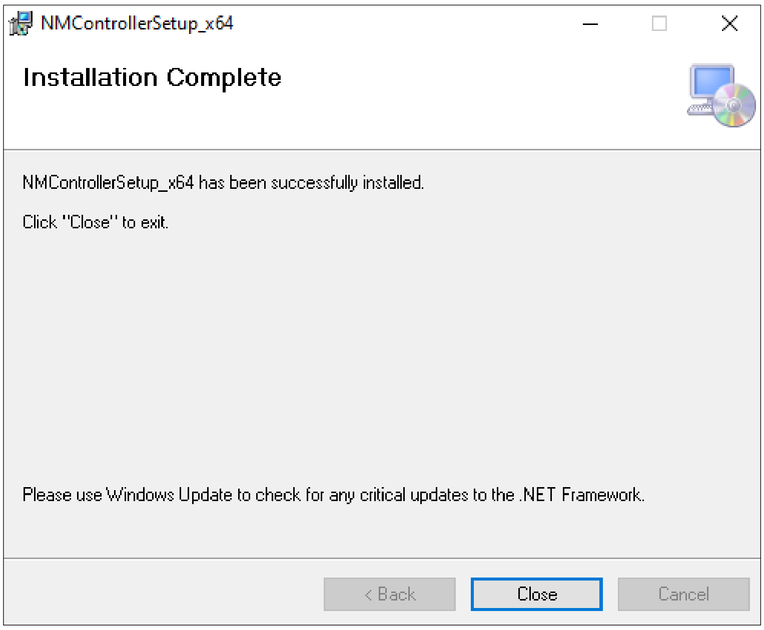
3. Click “**Next**”.



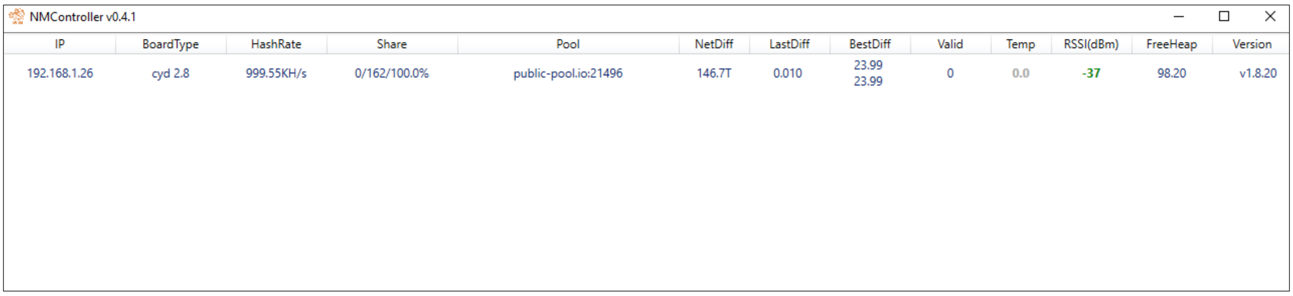
4. Click “**Browse**” to select the installation location, check “**Just me**”, and click “**Next**”.



5. Close the window when the installation completes.



Open the installed NMController Client and you can see the current device data.

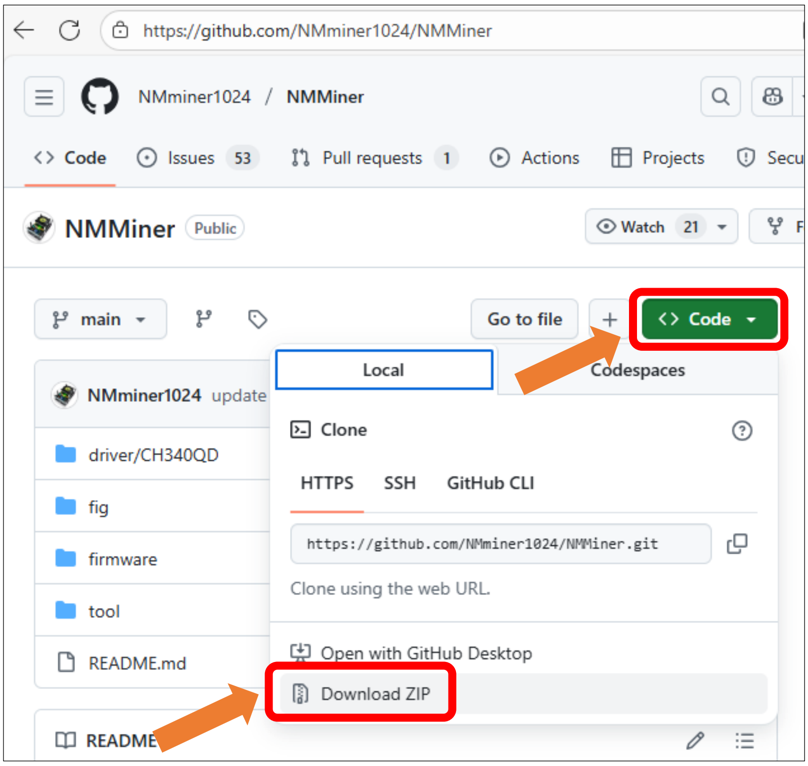


If you have any concerns, please feel free to contact us via support@freenove.com

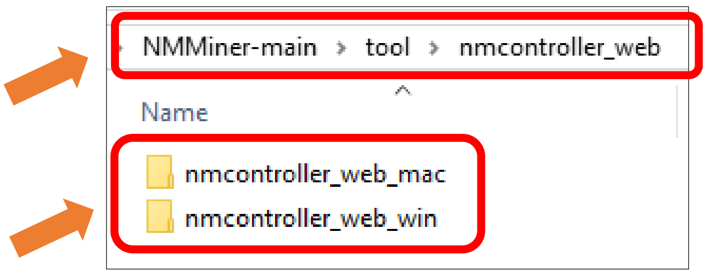
NMController Web

NMController Web: A cross-platform web client designed with Python, compatible with both Windows and macOS systems.

1. Click to open the NMMiner GitHub website, click **Code** -> **Download ZIP**



2. Extract the downloaded files, click **tool** -> **nmcontroller_web**, select the folder based on your computer's OS.



3. Double click to run the program.

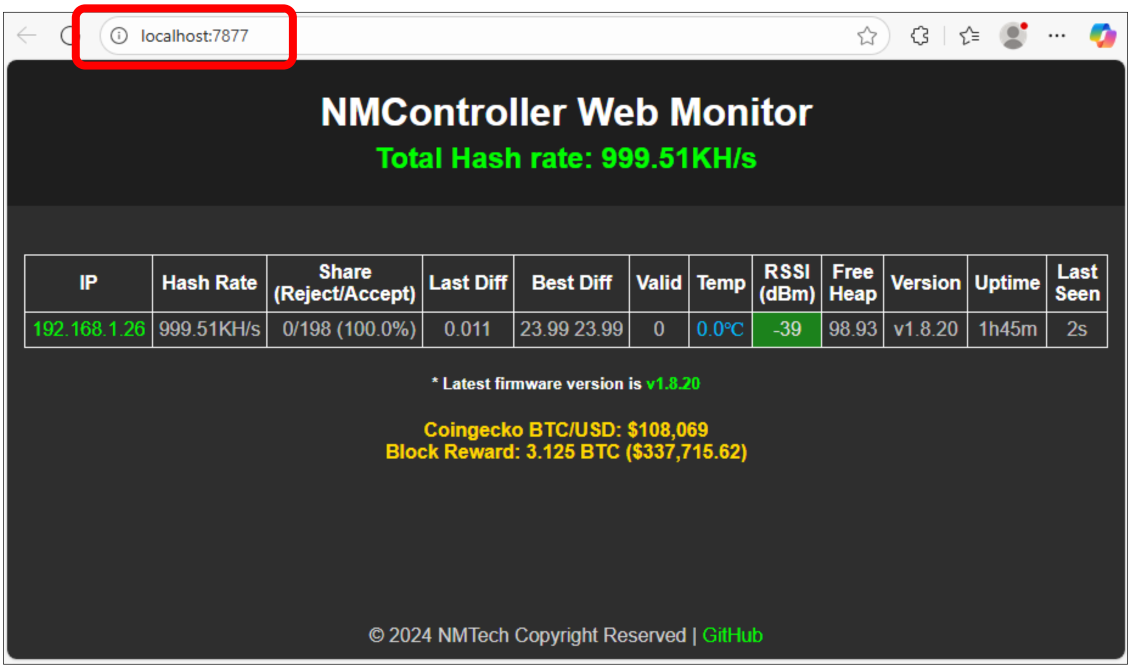
   (1) You can access the management interface by visiting **http://localhost:7877** on the current device.

   (2) Devices in the same LAN can access the management interface by visiting **http://192.168.1.24:7877**

Notes:

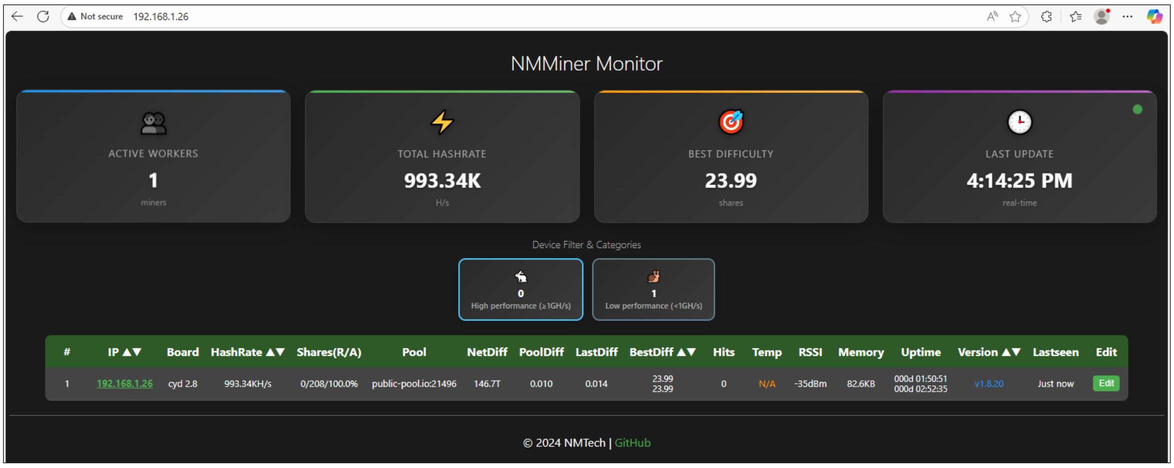
- **The following interface must NOT be closed when using NMController Web.**
- **The COM port may vary among devices. Please refer to the printed information for the correct port number.**

4. Access the corresponding address to open the management tool.



IP Address Management

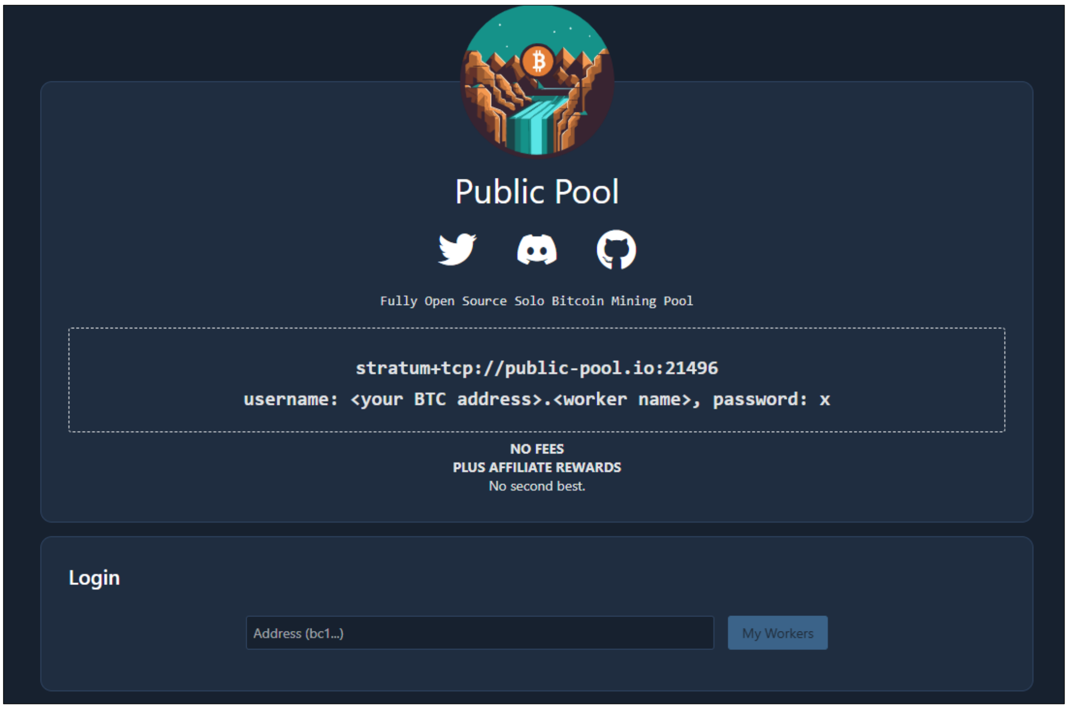
In the NMMiner interface, there is an IP address displayed. Devices connected to the same local network can access this IP address to open the management interface.



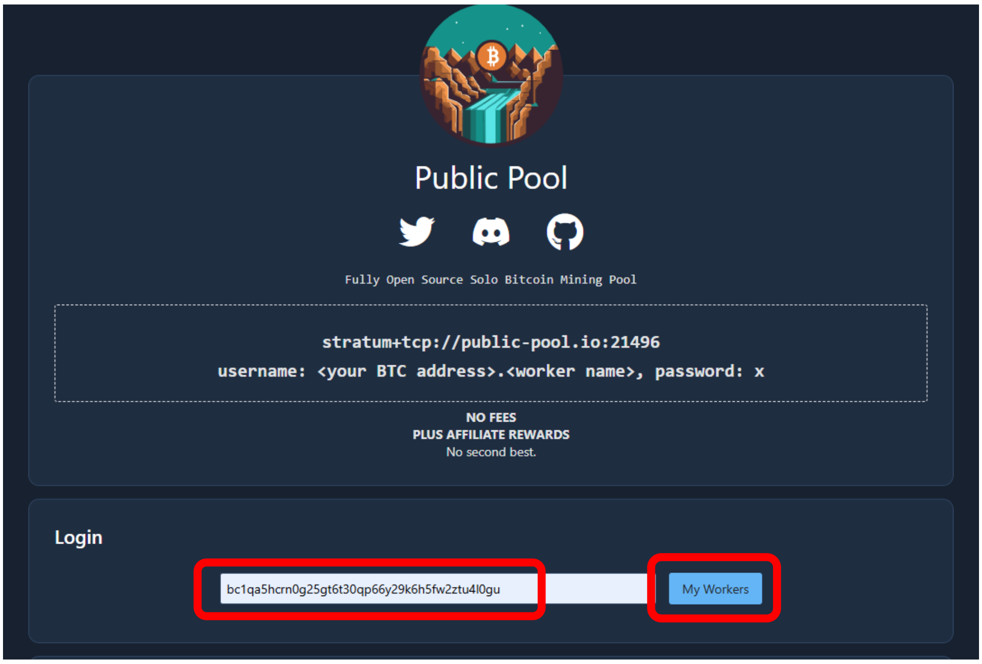
Public Pool

When the mining pool public-pool.io:21496 is selected, you can manage it via the web interface using the link (ttps://web.public-pool.io/#/).

1. Open public-pool



2. Enter BTC receiving address, and click "**My Workers**".



3. You can view the current working status of the worker.

Please note: This webpage only detects and manages miners currently working in the Public Pool that use your BTC address as the login or payout address. Once the pool is changed, this webpage will no longer be accessible for management.